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Department of Computer Science

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**An Integrated Smart Platform for Farm Solar Energy
Monitoring, Management & Grid Integration.**

Semester: Spring 25-26

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1. PROJECT PROPOSAL

1.1 Background to the Problem

In many farming areas especially in rural areas, farmers face problems because electricity is not always available. Fuel such as diesel is also very expensive. Because of this farmer's face difficulty in running irrigation systems, machines and other farming activities. This increases their cost and reduces their productivity.

At present, the world is moving toward clean and renewable energy. Solar energy is a good solution because it is clean, renewable and easily available from sunlight. Farmers can use solar panels to produce electricity. However, many farmers do not have a proper system to manage, monitor and maintain solar energy properly. Because of limited technical knowledge and support solar systems may not work efficiently for a long time.

To solve this problem, this project introduces **“An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.”** This system helps farmers produce, manage, store and use solar energy easily. Farmers can use this energy for their daily farming activities and reduce their dependency on diesel and other costly fuel sources.

When farmers produce extra energy, they can store it or send it to the national grid. By selling extra energy farmers can earn additional income. This also supports the use of renewable energy in the country. Stored energy can also be used during low sunlight or power outages.

The system allows farmers to see real-time information about energy production, energy consumption, energy storage and earnings. Government authorities can also monitor energy data and make better decisions. The system can help predict solar energy production using weather data. With farmer approval, extra energy can be transferred to the national grid. The system can also include a leaderboard and reward system to motivate farmers. It can also set energy production targets for farms.

Overall, this project helps farmers reduce cost, increase income, use clean energy and support the environment.

Group of User

This system is designed for different types of users:

Farm Owner: Farmers use the system to generate solar energy, monitor energy use, store energy, send extra energy to the grid and track their earnings.

Government Authority: Government authorities use the system to monitor energy production, set energy targets and manage rewards or incentives for farmers.

Energy Grid Operator: Energy grid operators manage the transfer of energy from farms to the national grid and make sure the grid works properly.

Investor: Investors use the system to view investment opportunities, analyze returns, monitor payments and track farmers earnings.

Admin / System Administrator: The admin controls the system, manages users, manages roles, and makes sure the system runs smoothly and securely.

Maintenance Team/Technician: Technicians monitor solar system performance, detect faults and perform maintenance or repair work to keep the system running properly.

This system helps Farmowner, Government authorities, Investors, Grid operators, Admins and Maintenance team work together to create a smart and sustainable energy solution.

1.2 Selection of Process Model

For this project, we have selected the **Incremental Development Model**.

Incremental Development Model is a software development model where the full system is not developed at one time. Instead, the system is divided into small parts called increments. Each increment contains analysis, design, coding, testing, and delivery. After completing one increment, the next increment is developed. In this way, the complete system is built step by step.

In our project, “**An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration,**” there are many different modules. These modules can be developed separately. For example, registration, login, profile management, energy monitoring, grid transfer, payment monitoring, admin panel, maintenance request, and report generation can be completed one by one.

Why We Chose Incremental Development Model?

We chose the Incremental Development Model because our project has many separate features and modules. The requirements are mostly clear and divided into parts such as user management, energy monitoring, grid management, payment, admin control, maintenance and reports. So, it is easier to develop the system step by step instead of developing everything at once.

In this model, when the development of one increment starts, the requirements of that increment are fixed. This keeps the current work stable. However, the requirements for later increments can still be changed or improved if needed.

This model also helps us test each module separately. If any problem is found, it can be fixed before moving to the next module. This reduces risk and makes the project easier to manage.

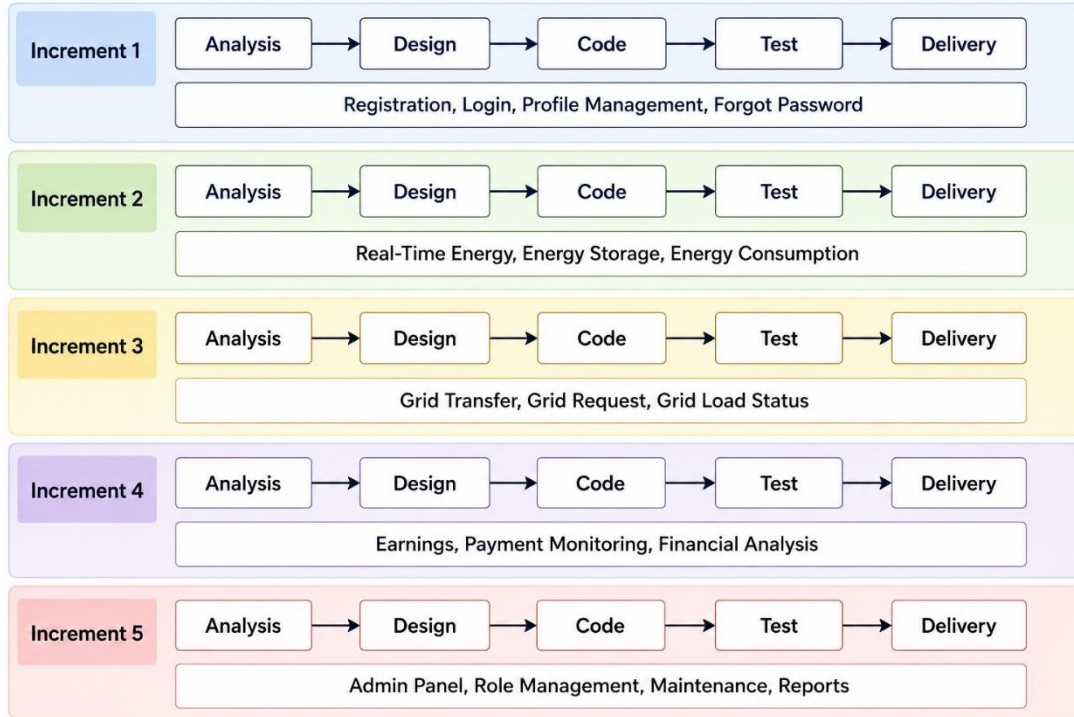


Fig 1: Incremental Development Model for the Proposed System.

We did not choose Waterfall because it is too strict and does not easily support later changes. We also did not choose Scrum, XP or DSDM because they need regular sprint meetings, strong customer involvement, pair programming, timeboxing, and continuous feedback. Since our project is an academic project based on PRD, diagrams, Gantt chart, test cases and planned modules. The Incremental Development Model is more suitable, simple and manageable.

2. Timeline Chart / Gantt Chart

$$\begin{aligned}
 \text{Effort} = PM &= \text{Coefficient}_{\langle \text{Effort Factor} \rangle} \times (\text{SLOC}/1000)^P \\
 &= 3.0 \times (25000/1000)^{1.12} \\
 &= 110.36 \text{ Person-Month}
 \end{aligned}$$

$$\begin{aligned}
\text{Development time} = DM &= 2.50 \times (PM)^T \\
&= 2.50 \times (110.36)^{0.35} \\
&= 12.97 \text{ Month}
\end{aligned}$$

$$\begin{aligned}
\text{Required number of people} = ST &= \frac{PM}{DM} \\
&= \frac{110.36}{12.97} \\
&= 8.48 \approx 9 \text{ Person}
\end{aligned}$$

Software Project Type	Coefficient (a)	P	T
Organic	2.4	1.05	0.38
Semi-detached	3.0	1.12	0.35
Embedded	3.6	1.20	0.32

Table 1: COCOMO Model Coefficients.

The project is considered a semi-detached software project because it has medium complexity, multiple modules, several user roles, database operations, real-time monitoring and external grid related features. Therefore, the semi-detached COCOMO values are used for effort estimation.

The Gantt chart shows the planned project timeline from February to July 2026. It includes project proposal, process model selection, requirement analysis, 32-functional requirements, 20-nonfunctional requirements, system design, UI/wireframe design, Trello board, Git workflow, software testing, bug fixing and final report submission.

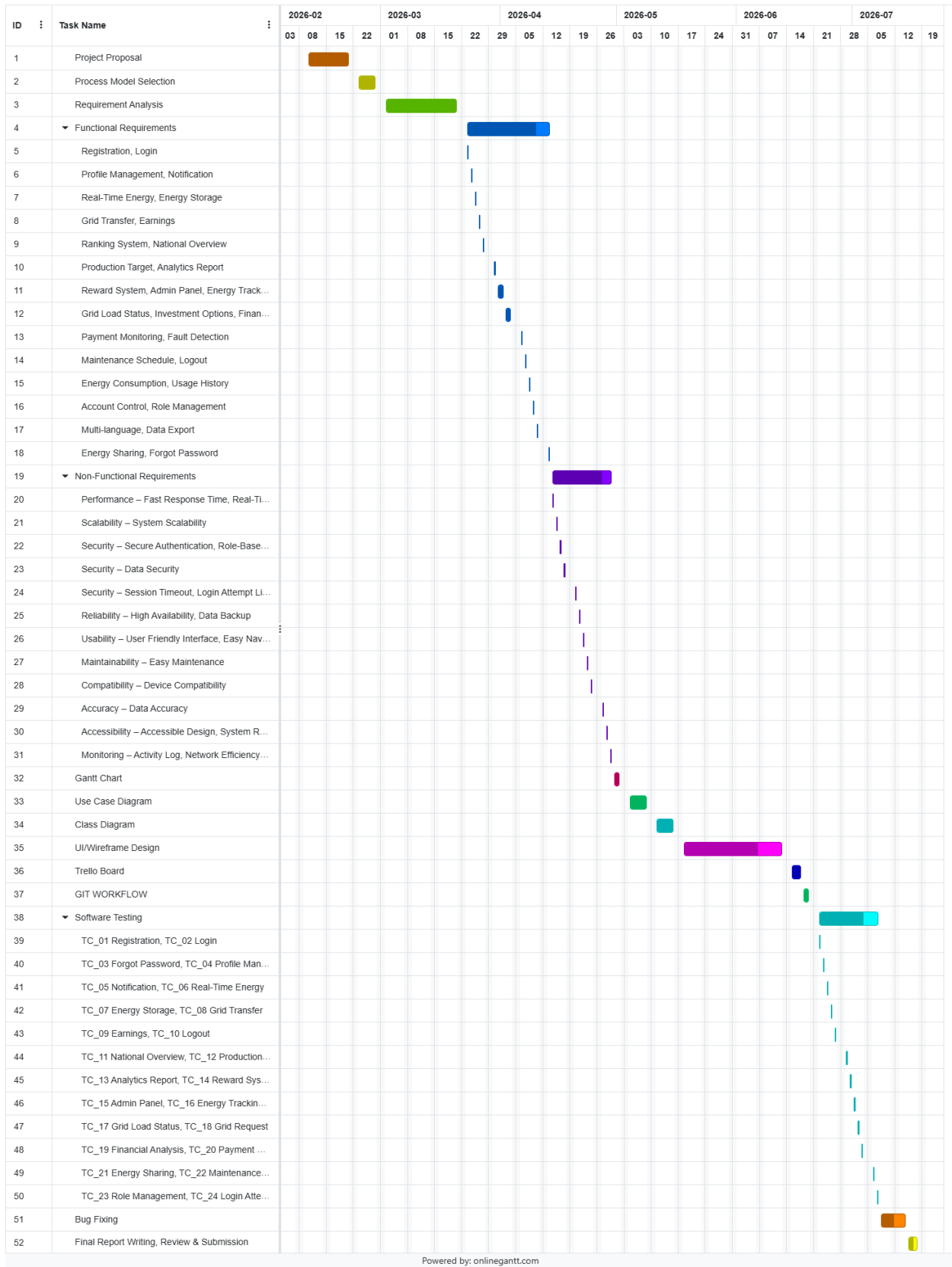


Fig 2: Gantt Chart for Smart Farm Solar Platform Project Timeline.

3. REQUIREMENTS DOCUMENT (PRD)

3.1 Functional Requirements

Item	Feature	Specification	Screen Definition	User Story	Acceptance Criteria
1	Registration	The system should allow users to create an account to access the platform.	Header: Platform logo and title “Create Account” Content: A registration form including full name, email, password, confirm password and role selection dropdown Actions: Signup button and login redirect link	As a Farm Owner/Government Authority/Energy Grid Operator/Investor/Admin/Maintenance Team, I should be able to register so that I can access the system.	- The system should validate all required fields - Email should be unique - Password should be ≥ 8 characters - Password and confirm password should match - Account should be created within 30 seconds - User should be redirected to the dashboard after successful registration.
2	Login	The system should allow users to Login to use their accounts securely.	Header: Title “Login to System” Content: A login form including email, password and remember me checkbox Actions: Login button and forgot password link	As a Farm Owner/Government Authority/Energy Grid Operator/Investor/Admin/Maintenance Team, I should be able to Login so that I can use my account.	- The system should verify credentials against the database - Successful login should redirect users to the dashboard - Invalid credentials should display an error message - Response time should be less than 2 seconds.
3	Profile Management	The system should allow users to update their	Header: Title “User Profile” Content: Profile form including name,	As a Farm Owner/Government Authority/Energy Grid	- The system should allow editing of profile fields - Data should be

		personal information to keep their profiles correct.	contact and password update option Actions: Save changes button	Operator/Investor/Admin/Maintenance Team, I should be able to manage my profile so that I can update and edit information.	validated before saving - Changes should be reflected immediately - Confirmation message should be displayed.
4	Notification	The system should provide real-time notifications to keep users updated about important events.	Header: Title “Notifications” Content: A list of notification items showing icon type (info/warning/critical), message text and timestamp Actions: Mark as read button	As a Farm Owner/Government Authority/Energy Grid Operator/Investor/Admin/Maintenance Team, I should be able to receive alerts so that I stay updated.	- Notifications should be delivered in real-time - Alerts should be categorized (warning, info, critical) - Notifications should be visible within 2 seconds of trigger
5	Real-Time Energy	The system should show live energy production to monitor performance instantly.	Header: Title “Energy Overview” Content: Energy production graph, live status indicator, and total energy generated summary Controls: Time range filter (daily/weekly/monthly)	As a Farm Owner/Government Authority, I should be able to view live energy production so that I can monitor output.	- The system should update data every 5 seconds - Graph should load within 2 seconds - Data accuracy $\geq$ 95% - Historical overlay should be available.
6	Energy Storage	The system should show stored energy levels to manage usage efficiently.	Header: Title “Energy Storage” Content: Battery percentage indicator and storage usage graph Alerts: Low storage warning	As a Farm Owner, I should be able to check stored energy so that I can manage usage.	- Battery percentage should be accurate - Low storage alert should be triggered below the threshold - Data refresh rate $\leq$ 5 seconds

7	Grid Transfer	The system should allow Farm Owners to send energy to the grid to earn money.	Header: Title “Grid Energy Transfer” Content: Input field for energy amount and grid connection selection Status: Transfer status and transaction history Actions: Submit transfer button	As a Farm Owner/Energy Grid Operator, I should be able to send energy to the grid and monitor and control energy transfer so that I can earn money and maintain grid stability.	- Transfer requests should be submitted successfully - Approval/rejection workflow should function correctly - Transfer status should be updated in real-time - Error handling should be available for failed transactions.
8	Earnings	The system should show income details to track earnings.	Header: Title “Earnings Overview” Content: Income graph and total earnings summary Controls: Time range filter	As a Farm Owner/Investor, I should be able to view income details so that I can track profit.	- Income should be calculated accurately. - Data should be updated in real-time. - Time filters should be available. - Graph should load within 2 seconds.
9	Ranking System	The system should rank farms based on performance to compare efficiency.	Header: Title “Performance Ranking” Content: Ranking table with columns for farm name, production, and rank Controls: Sorting by production	As a Government Authority, I should be able to compare farm performance so that I can evaluate efficiency.	- Rankings should be updated daily. - Sorting functionality should be available.
10	National Overview	The system should show national energy data to help the government plan better policies.	Header: Title “National Energy Overview” Content: Total energy production chart, energy distribution map, and country statistics summary	As a Government Authority, I should be able to view national energy data so that I can plan policies.	- Data should be collected from all farms. - Dashboard should load within 3 seconds.

			Controls: Region filter		- Charts and graphs should show correct information.
11	Production Target	The system should allow setting energy targets to help farms follow goals.	Header: Title “Production Target” Content: Target value input and target vs actual chart Actions: Save target button	As a Government Authority, I should be able to set targets so that I can help farms follow energy goals.	- Target values should be saved successfully. - Targets should be visible to Farm Owners. - Data should be editable by authorized users.
12	Analytics Report	The system should generate reports to analyze performance.	Header: Title “Analytics Report” Content: Performance metric charts and data summary Actions: Export report button	As a Government Authority, I should be able to generate reports so that I can analyze system performance.	- Reports should be generated correctly. - Export option should be available. - Load time should be less than 3 seconds.
13	Reward System	The system should provide rewards to keep users motivated.	Header: Title “Reward Management” Content: List of eligible users and reward details Actions: Assign reward button	As a Farm Owner/Government Authority, I should be able to receive and assign rewards so that I stay motivated and encourage Farm Owners.	- Rewards should be assigned correctly. - Notification should be sent to the user.
14	Admin Panel	The system should allow admins to control and manage the entire system.	Header: Title “Admin Dashboard” Navigation: User menu, role menu, and system settings menu Content: Control panel	As an Admin, I should be able to control the system so that I can ensure proper system operation.	- All controls should function correctly.
15	Energy Tracking Map	The system should visualize	Header: Title “Energy Flow Map” Content: Map	As an Energy Grid Operator/Government Authority, I should be	- Map should update in real-time. - Energy flow should be clearly visible.

		real-time energy flow and direction on a map to track distribution effectively.	interface with energy flow layer, farm location markers, and flow direction animation Controls: Zoom controls and refresh button	able to track energy flow so that I can monitor distribution.	- Data should refresh within 5 seconds.
16	Grid Load Status	The system should show grid load status to help operators balance energy.	Header: Title “Grid Load Status” Content: Grid load graph and current load indicator Alerts: Overload warning	As an Energy Grid Operator, I should be able to check grid load so that I can balance energy.	- Real-time load data should be displayed. - Load thresholds should be visualized. - Alerts should be triggered on overload conditions.
17	Investment Options	The system should show investment opportunities to help investors choose projects.	Header: Title “Investment Opportunities” Content: List of project cards including project name and expected return Actions: View details button	As an Investor, I should be able to view projects so that I can make investment decisions.	- Project list should load within 2 seconds. - Details should be clearly displayed.
18	Financial Analysis	The system should analyze return on investment to help investors make decisions.	Header: Title “Financial Analysis” Content: ROI graph and expected return summary Controls: Investment amount input	As an Investor, I should be able to analyze return on investment so that I can make decisions.	- ROI calculations should be accurate - Data should be updated dynamically.
19	Payment Monitoring	The system should track payments to keep	Header: Title “Payment Monitoring” Content: Payment history table with date, amount, and status	As a Farm Owner/Investor/Admin, I should be able to track payments so	- Payment data should be accurate - The Table should support sorting and filtering.

		financial data clear.	Controls: Transaction filter and search bar	that I can maintain clear financial data.	- Transactions should be updated in real-time - The system should display the correct date, amount and status.
20	Fault Detection	The system should detect system faults to fix issues quickly.	Header: Title “System Fault Alerts” Content: List of fault notifications with severity indicator Actions: View details button	As a Farm Owner/Maintenance Team, I should be able to detect and identify faults so that I can report and fix issues quickly.	- Faults should be detected automatically. - Alerts should be triggered instantly.
21	Maintenance Schedule	The system should schedule maintenance tasks to keep system operations smooth.	Header: Title “Maintenance Schedule” Content: Task schedule calendar Actions: Add task button	As a Maintenance Team/Farm Owner, I should be able to schedule and view maintenance tasks so that maintenance operations remain organized and service schedules are known.	- Tasks should be saved successfully. - Tasks should be visible to relevant users.
22	Logout	The system should allow users to log out to keep their accounts secure.	Header: Title “Logout” Actions: Logout button	As a Farm Owner/Government Authority/Energy Grid Operator/Investor/Admin/Maintenance Team, I should be able to log out so that I can keep my account secure.	- Session should be terminated. - Users should be redirected to the login page.
23	Energy Consumption	The system should show energy usage	Header: Title “Energy Consumption” Content: Consumption graph	As a Farm Owner, I should be able to track energy usage so	- Data accuracy should be greater than or equal to 95%.

		to optimize consumption.	Controls: Time range filter	that I can optimize energy consumption.	- Real-time updates should be available.
24	Usage History	The system should provide past data to analyze trends.	Header: Title “Usage History” Content: Historical data table Controls: Date range filter	As a Farm Owner/Government Authority/Investor, I should be able to view past data so that I can analyze trends.	- Data should load correctly. - Filters should function properly.
25	Account Control	The system should allow admins to manage user accounts to maintain security.	Header: Title “User Account Management” Content: User list Actions: Enable/disable user button	As an Admin, I should be able to manage accounts so that I can maintain system security.	- Option should be available for enable/disable accounts. - Changes should be reflected instantly.
26	Role Management	The system should allow role assignments to maintain access control.	Header: Title “Role Management” Content: Role assignment dropdown Actions: Update role button	As an Admin, I should be able to assign roles so that I can maintain access control.	- Role updates should be applied immediately.
27	Multi-language	The system should support multiple languages to help users understand easily.	Header: Title “Language Settings” Content: Language toggle option	As a Farm Owner/Government Authority/Energy Grid Operator/Investor/Admin/Maintenance Team, I should be able to select local languages so that I can understand the system easily.	- Language should switch instantly.
28	Data Export	The system should allow data export to use exported	Header: Title “Export Data” Actions: Download PDF and CSV buttons	As a Government Authority/Investor/Admin, I should be able to export data so	- Files should be downloadable in PDF/CSV format. - Exported data

		data for analysis.		that I can perform analysis.	should match system records.
29	Maintenance Request	The system should allow users to report issues to resolve problems.	Header: Title “Maintenance Request” Content: Issue submission form including description and image upload Actions: Submit request button	As a Farm Owner, I should be able to request maintenance so that I can resolve issues.	- Requests should be submitted successfully. - Status tracking should be available.
30	Grid Request	The system should allow users to request grid connection to sell energy.	Header: Title “Grid Connection Request” Content: Request submission form Actions: Submit request button Status: Approval status	As a Farm Owner, I should be able to request grid connections so that I can sell energy.	- Approval/rejection workflow should work correctly.
31	Energy Sharing	The system should allow farm owners to share energy with nearby farms.	Header: Title “Energy Sharing” Content: Input field for energy amount and recipient selection Actions: Share energy button	As a Farm Owner, I should be able to share energy so that nearby farms benefit.	- Sharing process should complete successfully.
32	Forgot Password	The system should allow users to reset their password to regain access.	Header: Title “Forgot Password” Content: Password reset form with email input Actions: Send reset link button and back to login link	As a Farm Owner/Government Authority/Energy Grid Operator/Investor/Admin/Maintenance Team, I should be able to reset passwords so that I can regain account access if forgotten.	- The system should validate the entered email - Email should exist in the system - Reset link should be sent within 30 seconds - Reset link should be secure and time-limited - User should be able to set a new password - Password should be

					≥ 8 characters - Password update should be successful - User should be redirected to the login page after reset.
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3.2 Non-Functional Requirements

Item	Feature	Specification	Screen Definition	User Story	Acceptance Criteria
1	Performance - Fast Response Time	The system should provide fast response time so that users should use the system smoothly.	Header: Title “System Performance” Content: All system screens Actions: System response monitoring	All users should receive fast system response so that they should use the system smoothly.	- The system should load pages within 2 seconds - The system should respond to user actions without delay - The system should maintain consistent performance
2	Performance - Real-Time Data Update	The system should update data in real time so that users should monitor the system instantly.	Header: Title “Live Data Update” Content: Energy dashboard Controls: Refresh monitoring	All users should receive real-time data updates so that they should monitor energy instantly.	- The system should update data every 5 seconds - The system should display latest data without manual refresh - The system should ensure minimal delay in updates
3	Scalability-System Scalability	The system should support multiple users and future growth without performance issues.	Header: Title “System Usage” Content: All system pages Controls: Load monitoring	All users should use the system during peak time without performance issues.	- The system should support multiple users - The system should not crash under high load - The system should be Stable under load.
4	Security - Secure Authentication	The system should ensure secure logins for all users.	Header: Title “Login Security” Content: Email & password input	All users should access the system through secure login so that their	- The system should allow access only to valid users - The system should

			Actions: Login validation	accounts should stay safe.	display an error for invalid login - The system should securely store passwords
5	Security - Role-Based Authorization	The system should restrict access based on roles.	Header: Title “Access Control” Content: System pages Controls: Role permissions	All users should access only authorized features through role-based access control.	- The system should restrict access based on roles - The system should block unauthorized access - The system should update permissions instantly
6	Security -Data Security	The system should protect sensitive data.	Header: Title “Data Security” Content: Sensitive data Actions: Data protection	All users should have protected and secure data.	- The system should protect sensitive data - The system should prevent unauthorized access. - The system should store data securely
7	Security - Session Timeout	The system should log out inactive users automatically.	Header: Title “Session Management” Content: Session activity Actions: Auto logout	All users should be logged out automatically after inactivity so that their accounts should stay safe.	- The system should log out users after inactivity. - The system should end sessions automatically - The system should redirect users to login page
8	Security - Login Attempt Limit	The system should limit login attempts for security.	Header: Title “Login Security Control” Content: Login attempts Actions: Attempt control	Login attempts should be limited so that system security should be ensured.	- The system should allow a maximum of 3 failed attempts - The system should block accounts after the limit is reached. - The system should notify users.
9	Reliability-High Availability	The system should remain available most of the time.	Header: Title “System Availability” Content: Uptime	All users should access the system whenever needed	- The system should remain available most of the time - The system should

			status Controls: Monitoring	without interruption.	minimize downtime - The system should recover quickly after failure
10	Reliability-Data Backup	The system should support data backup and recovery.	Header: Title “Backup System” Content: Backup storage Actions: Backup control	System data should be recoverable through backup management if needed.	- The system should back up data regularly - The system should allow data recovery - The system should prevent data loss
11	Performance-Resource Utilization	The system should use CPU, memory, and storage efficiently during operation.	Header: Title “Resource Usage” Content: System resource status Controls: Resource monitoring	System resources should be utilized efficiently so that the system should run smoothly.	-The system should optimize resource usage -The system should avoid unnecessary processing -The system should maintain stable operation
12	Usability-User Friendly Interface	The system should provide a simple interface.	Header: Title “User Interface” Content: All screens Controls: UI interaction	All users should use a simple and user-friendly interface without confusion.	- The system should be easy to understand - The system should require no training
13	Usability-Easy Navigation	The system should allow easy navigation so that users should find features quickly.	Header: Title “Navigation” Content: Menu and pages Controls: Navigation options	All users should navigate the system easily so that they should find features quickly.	-The menu should be simple and easy to understand -Users should find features quickly -All pages should look similar and consistent
14	Maintainability -Easy Maintenance	The system should be easy to maintain and update.	Header: Title “Maintenance System” Content: Maintenance tools Actions: Updates	The system should be easy to maintain so that problems should be fixed and updates should be applied quickly.	- The system should allow quick updates - The system should support bug fixing - The system should run during maintenance

15	Compatibility -Device Compatibility	The system should work on multiple devices.	Header: Title “Device Compatibility” Content: Device screens Controls: Layout adjust	All users should access the system from different devices.	- The system should work on all devices - The system should adjust to screen sizes - The system should maintain layout consistency
16	Accuracy - Data Accuracy	The system should ensure accurate data.	Header: Title “Data Accuracy” Content: System data Controls: Validation	All users should receive accurate data so that decisions should be correct.	- The system should display correct data - The system should prevent data mismatch - The system should update data properly
17	Monitoring - Activity Log	The system should track system activities.	Header: Title “Activity Log” Content: Logs Controls: Filtering	System activities should be tracked through activity logs.	- The system should record all actions - The system should include user and timestamp - The system should allow access to logs by authorized users
18	Accessibility - Accessible Design	The system should support accessible design.	Header: Title “Accessibility” Content: Accessible UI Controls: Display settings	All users should use an accessible system comfortably.	-The system should provide readable text -The system should ensure accessible design -The system should support proper color contrast
19	System Recovery	The system should handle failures and recover quickly without affecting system operations.	Header: Title “System Recovery” Content: System status indicator and recovery logs Controls: Refresh and system status monitoring	The system should recover from failures quickly so that system operations should continue smoothly.	- The system should recover within 5 seconds after failure - No data should be lost during system failure - The system should resume normal operation automatically

20	Network Efficiency- Low Internet Support	The system should work properly with a low internet connection.	Header: Title “Network Support” Content: System connection status Controls: Network monitoring	All users should use the system with low internet connectivity in rural areas.	-The system should work under low bandwidth -The system should reduce unnecessary data loading -The system should keep important features usable
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4. SOFTWARE DESIGN

4.1 System Design:

Use Case Diagram

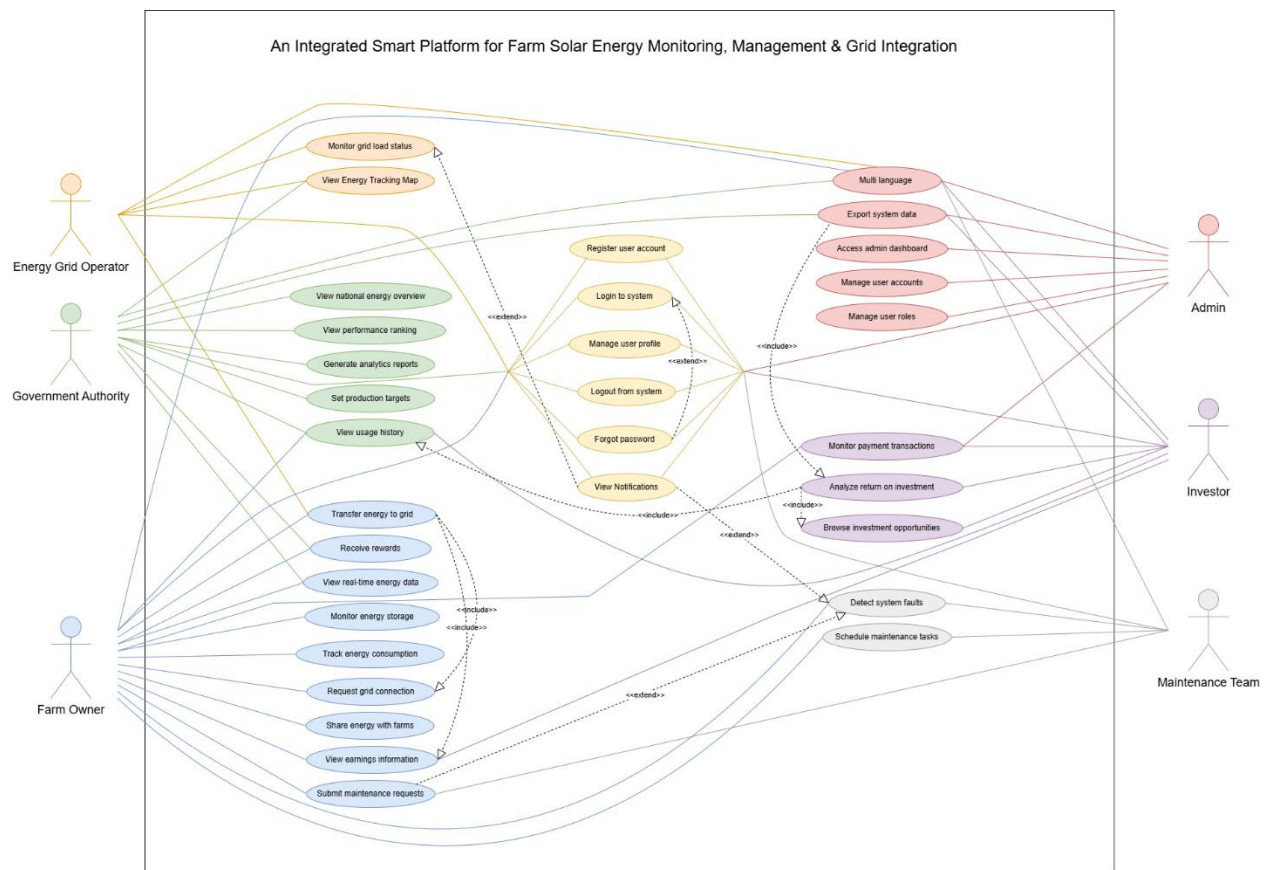


Fig 3: Use Case Diagram for Smart Farm Solar Platform System.

Class Diagram

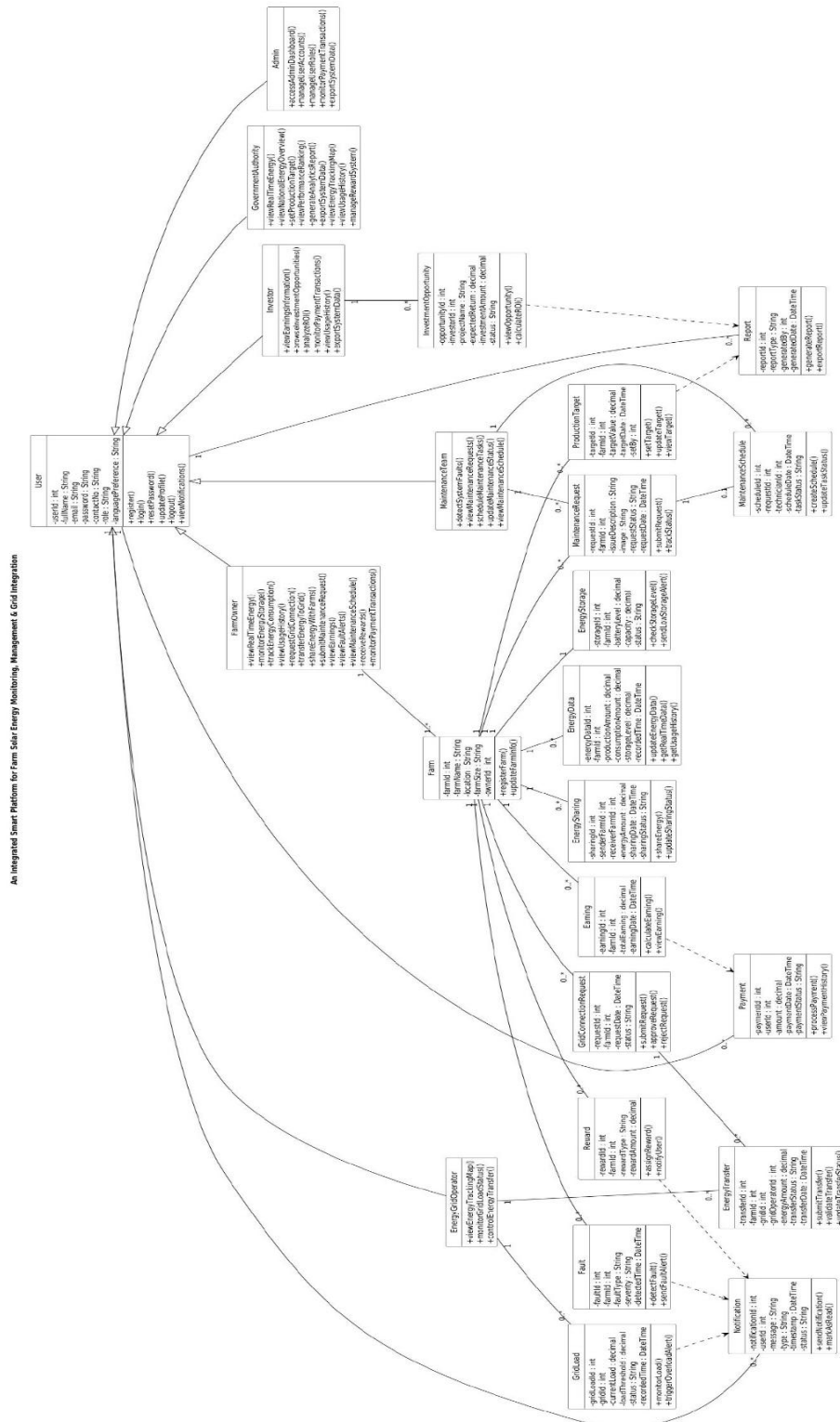
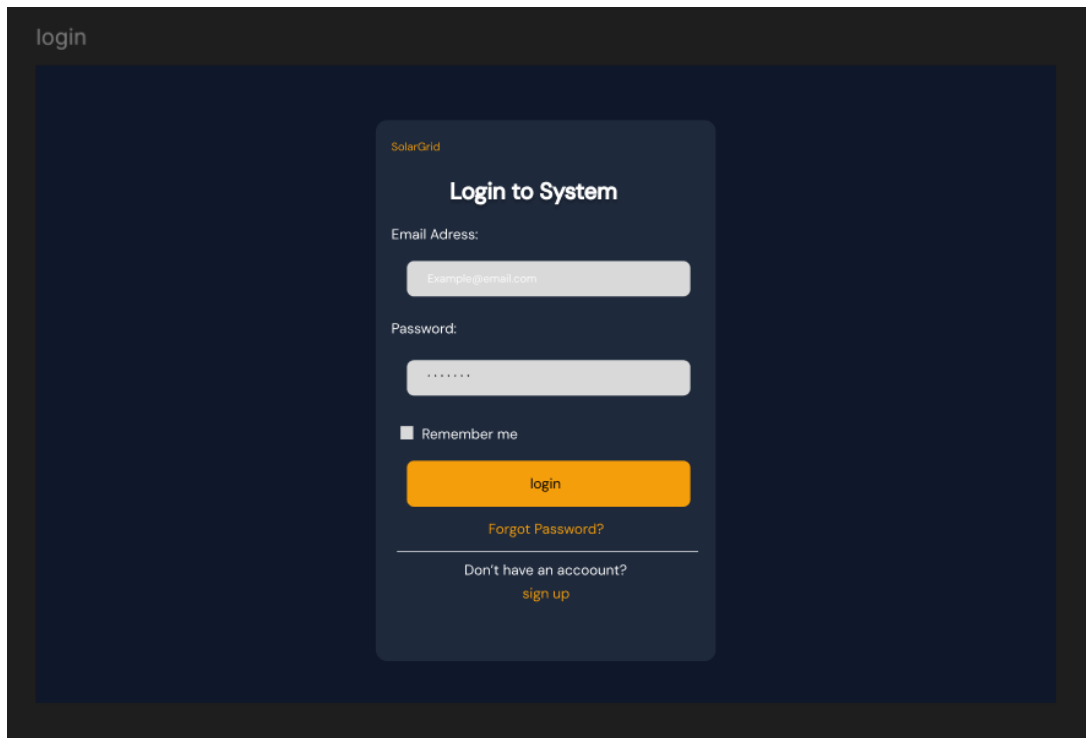


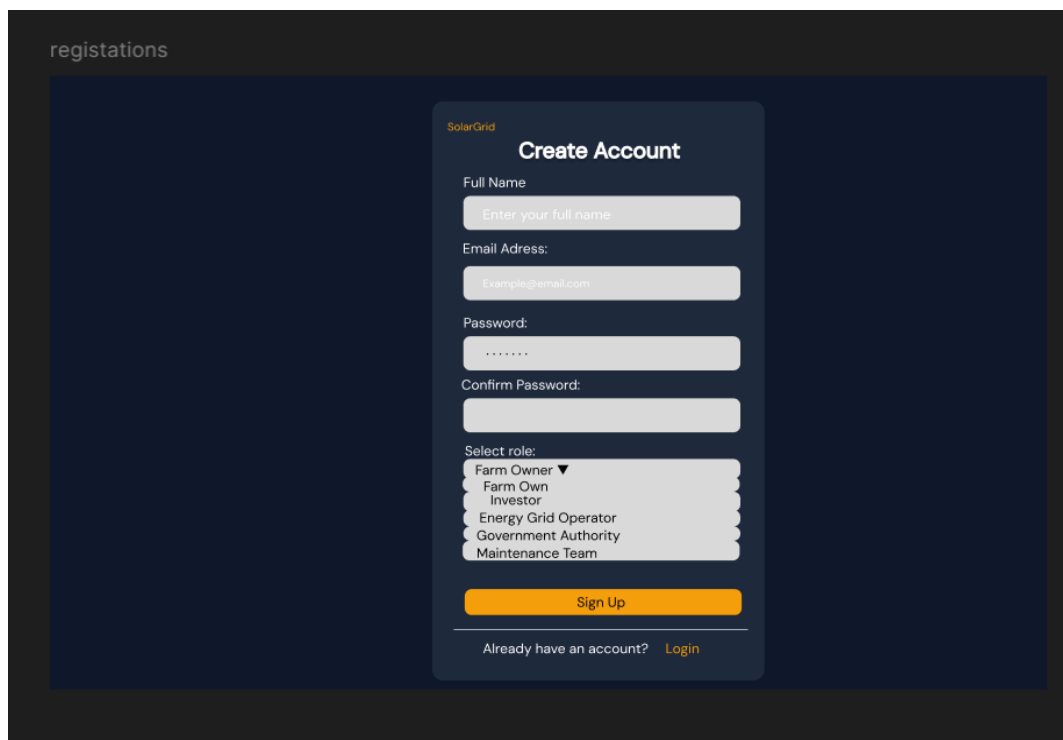
Fig 4: Class Diagram.

4.2 UI / Wireframe Design using Figma



The wireframe shows a login page with a dark blue background. At the top left, the word "login" is written in a small, light font. The main content is a light blue rounded rectangle. Inside, the "SolarGrid" logo is at the top left. The title "Login to System" is centered. Below it, the "Email Address:" label is followed by a text input field containing "Example@email.com". The "Password:" label is followed by a password input field with masked characters. Below the password field is a "Remember me" checkbox. A large orange "login" button is centered. Below the button is a "Forgot Password?" link. A horizontal line separates the login section from the registration section. Below the line, the text "Don't have an account?" is followed by a "sign up" link.

Fig 5: User Login Interface.



The wireframe shows a registration page with a dark blue background. At the top left, the word "registrations" is written in a small, light font. The main content is a light blue rounded rectangle. Inside, the "SolarGrid" logo is at the top left. The title "Create Account" is centered. Below it, the "Full Name" label is followed by a text input field containing "Enter your full name". The "Email Address:" label is followed by a text input field containing "Example@email.com". The "Password:" label is followed by a password input field with masked characters. The "Confirm Password:" label is followed by a text input field. Below the password fields is a "Select role:" label followed by a dropdown menu. The dropdown menu is open, showing a list of roles: "Farm Owner" (selected), "Farm Own", "Investor", "Energy Grid Operator", "Government Authority", and "Maintenance Team". Below the dropdown is a large orange "Sign Up" button. Below the button is a horizontal line. Below the line, the text "Already have an account?" is followed by a "Login" link.

Fig 6: User Registration Interface.

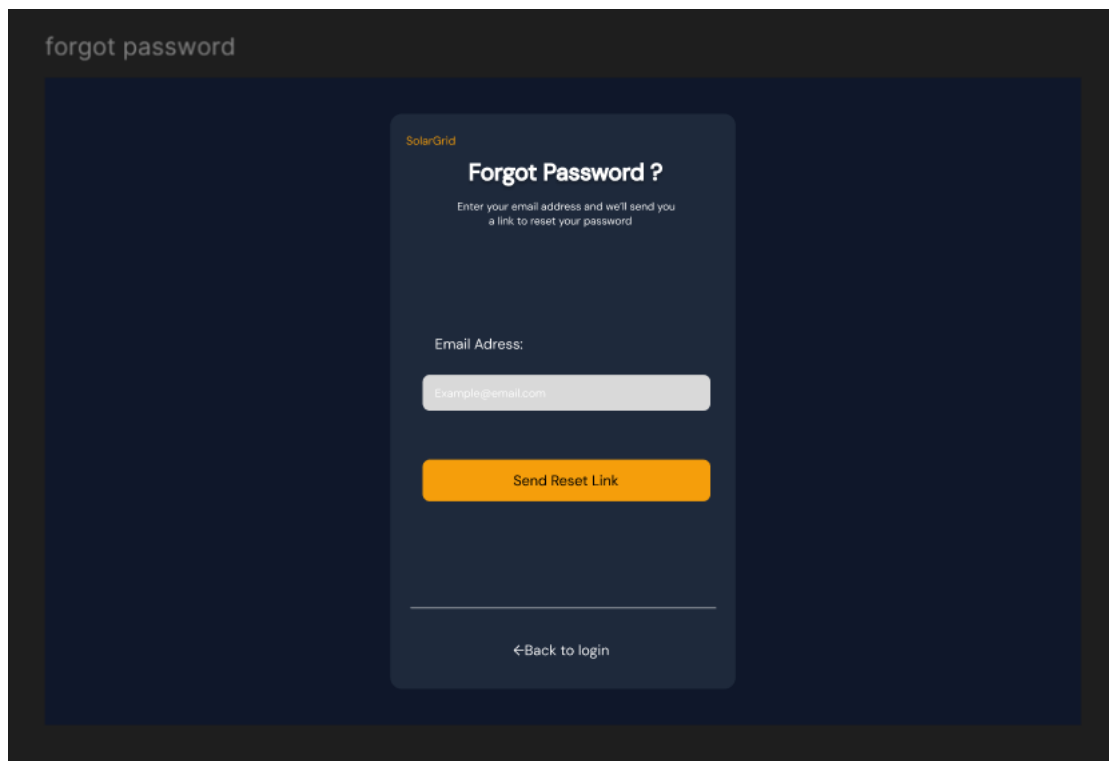


Fig 7: Forgot Password Interface.

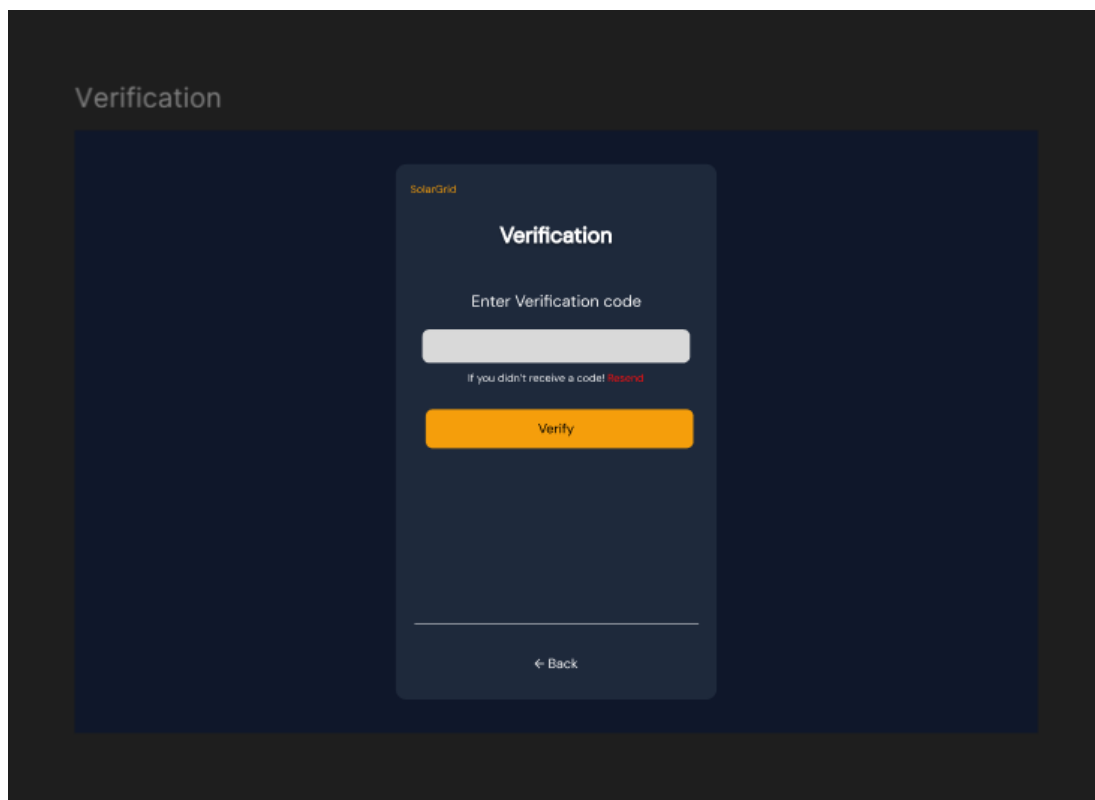


Fig 8: Verification.

new password

SolarGrid

New Password

Enter New Password

8 symbols at least

Confirm Password

.....

submit

Fig 9: Set New Password.

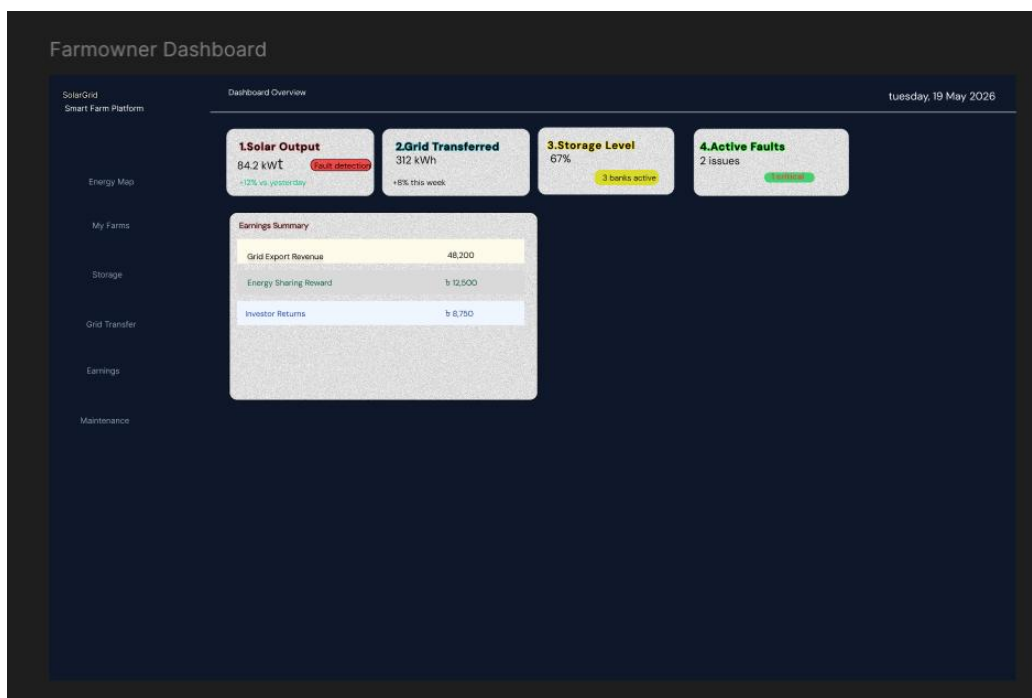


Fig 10: Farm Owner Dashboard Overview.



Fig 11: Dashboard to energy map .

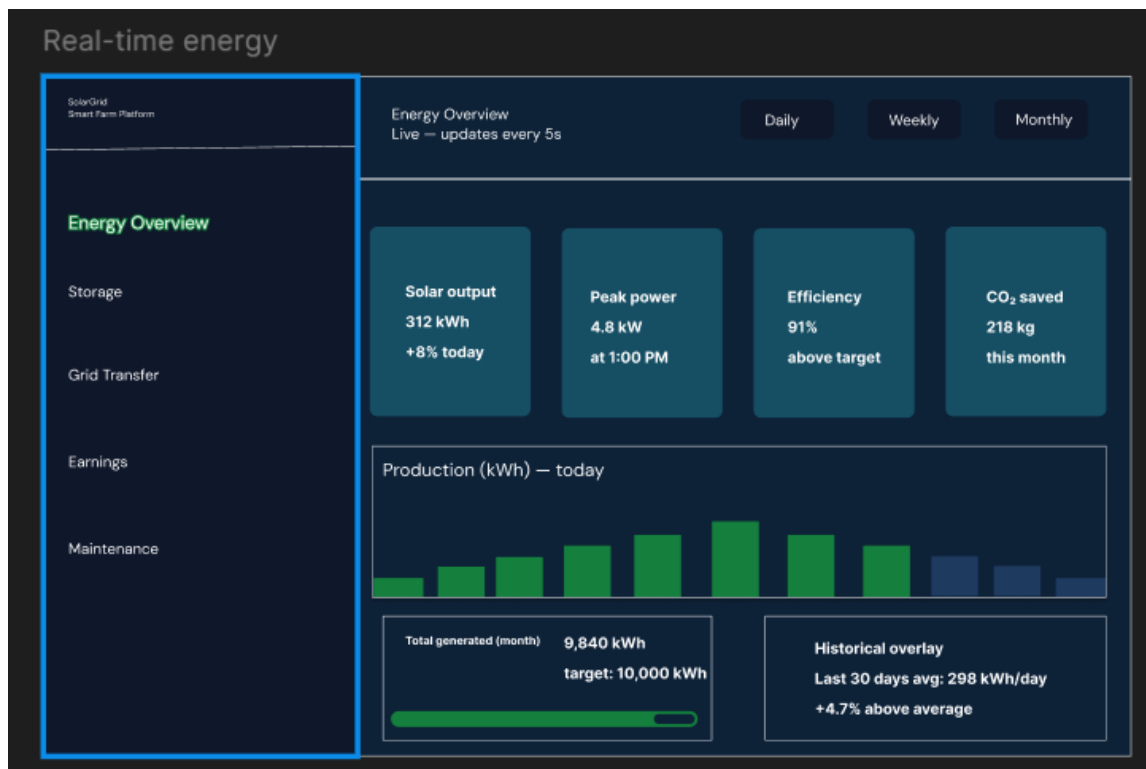


Fig 12: Real-Time Energy Monitoring Dashboard

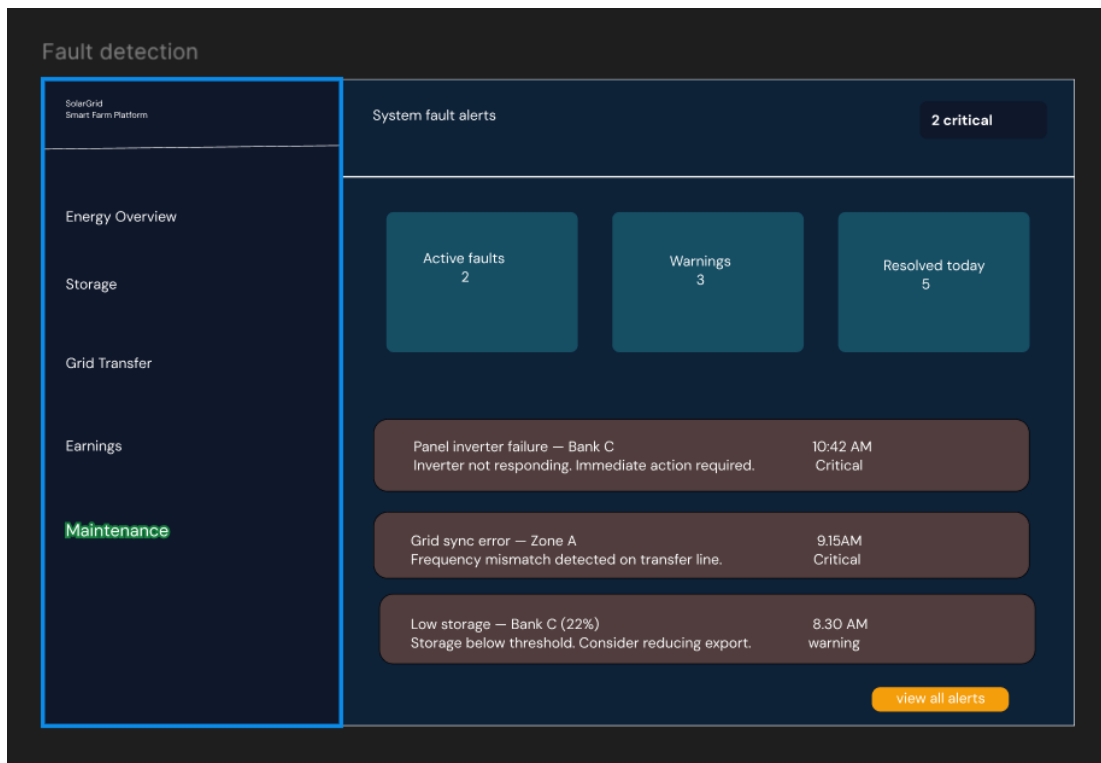


Fig 13: Fault Detection Dashboard.

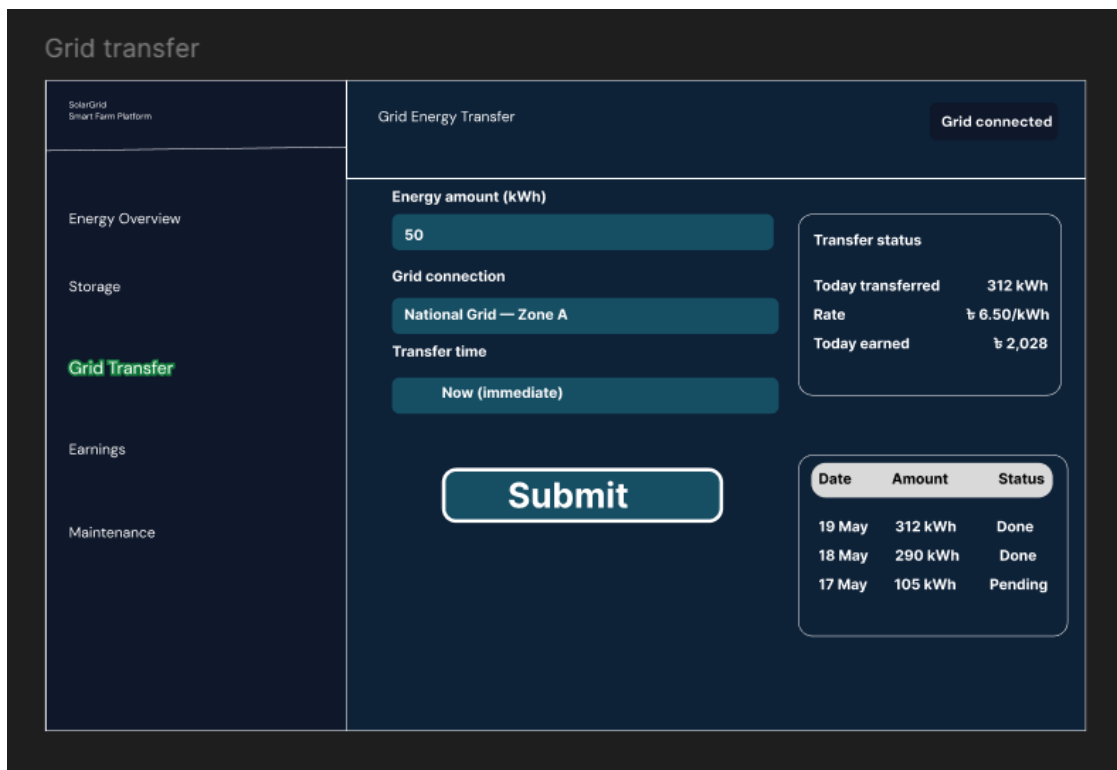


Fig 14: Grid Energy Transfer Interface

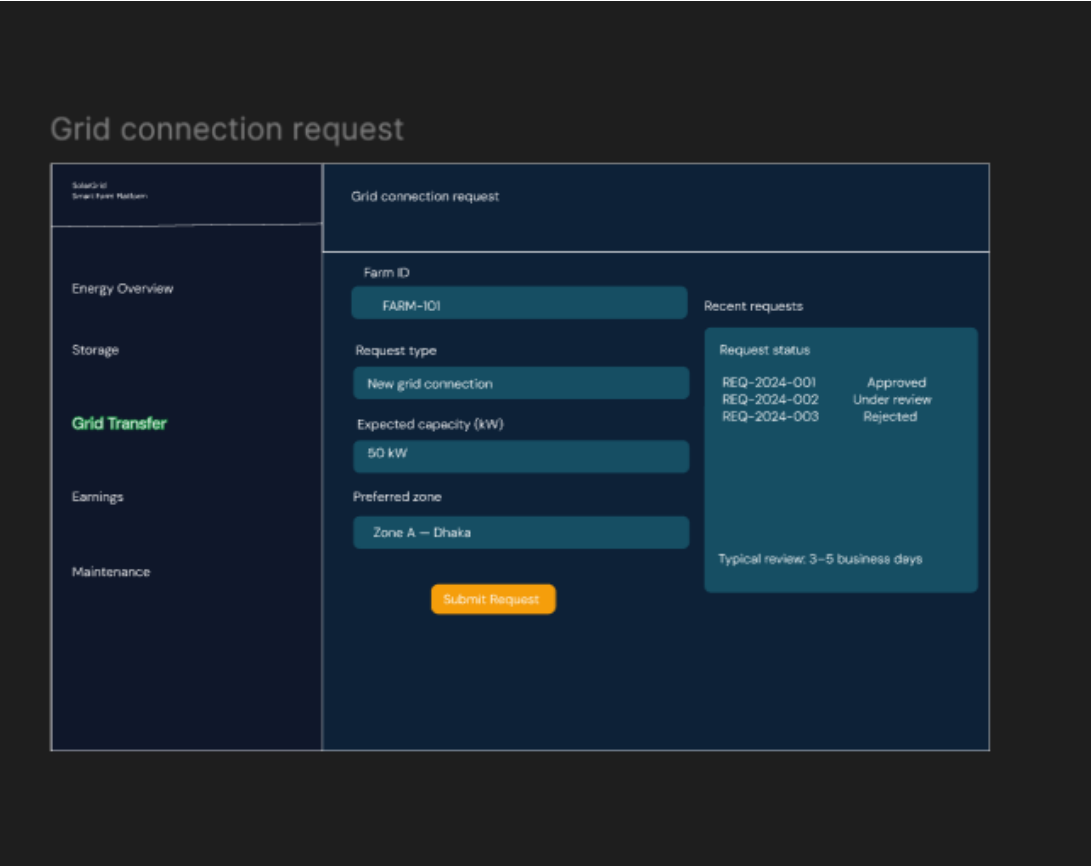


Fig 15: Grid Connection Request Interface.

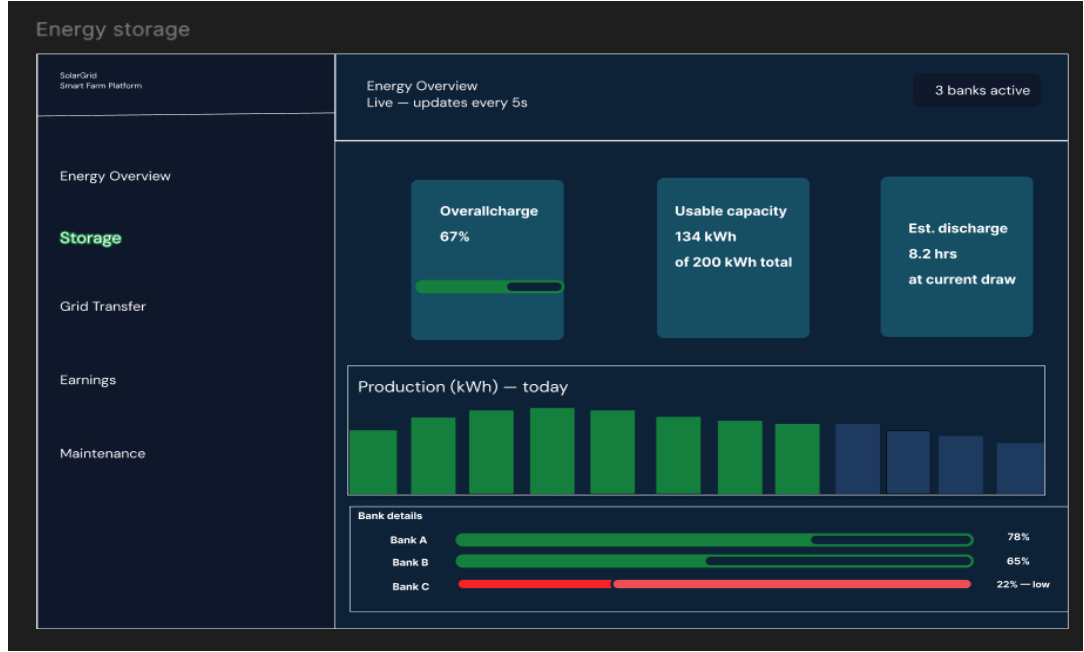


Fig 16: Dashboard to Storage.

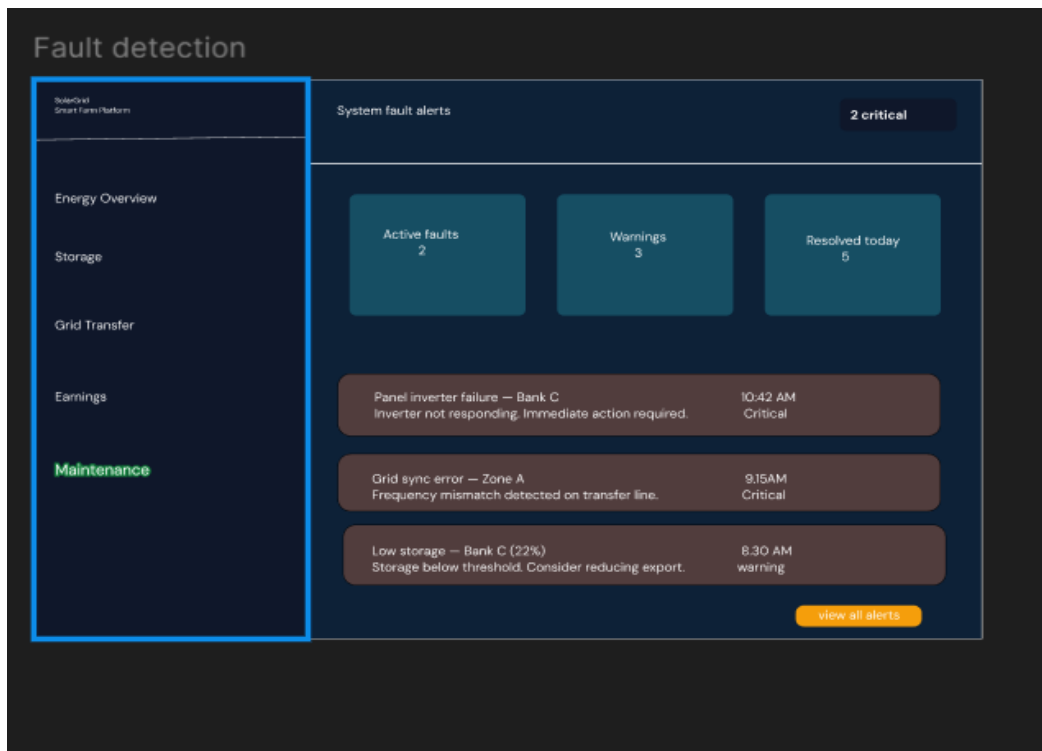


Fig 17: Fault Detection.

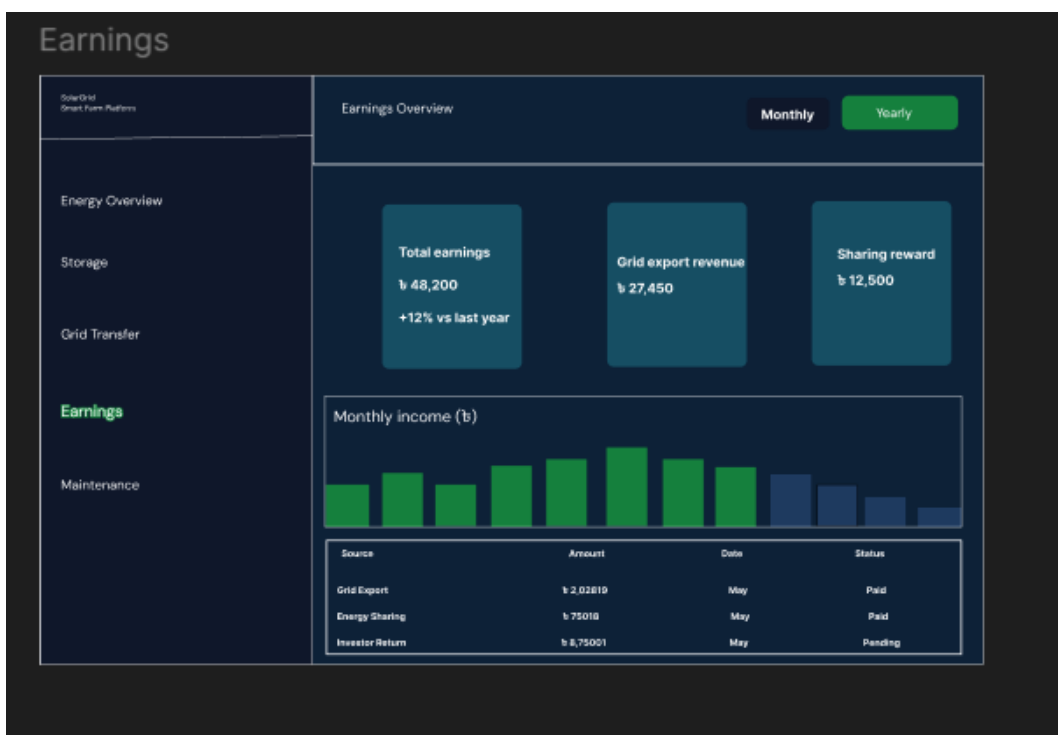


Fig 18: Earning interface.

SolarGrid
Smart Farm Platform

Energy Overview

Storage

Grid Transfer

Earnings

Maintenance

Maintenance request

2 critical

Issue description

Solar panel not working on row 3...

Priority

High — urgent

Upload image

Click to upload image

Submit

Recent requests

Issue	Date	Status
Panel damage	15 May	Fixed
Inverter fault	10 May	Open
Wiring issue	02 May	Fixed

Fig 19:Maintance.

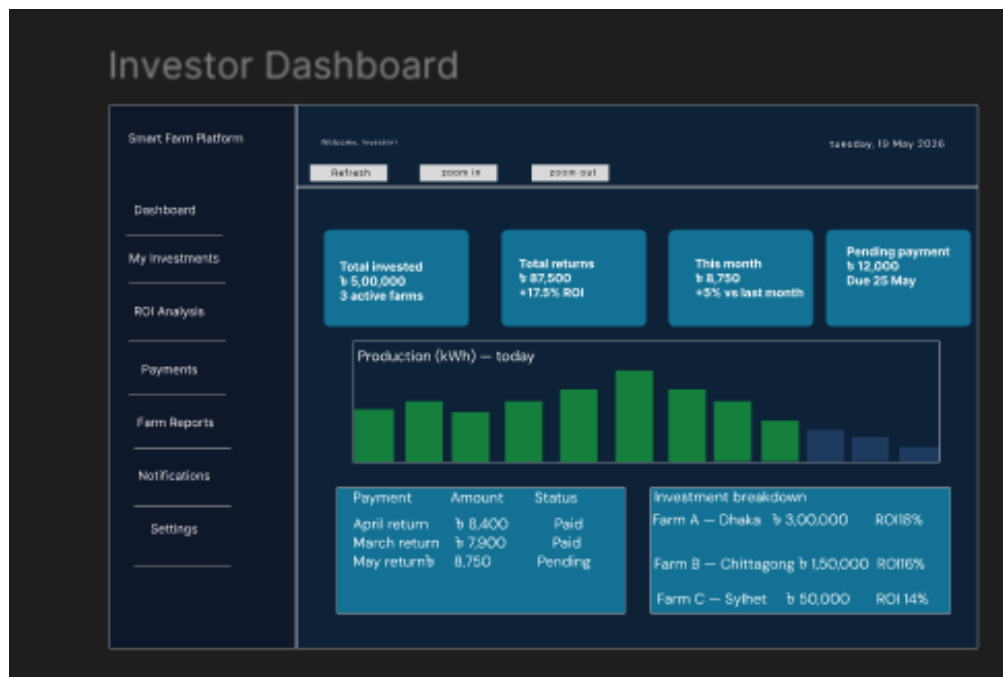


Fig 20: Investor Dashboard.

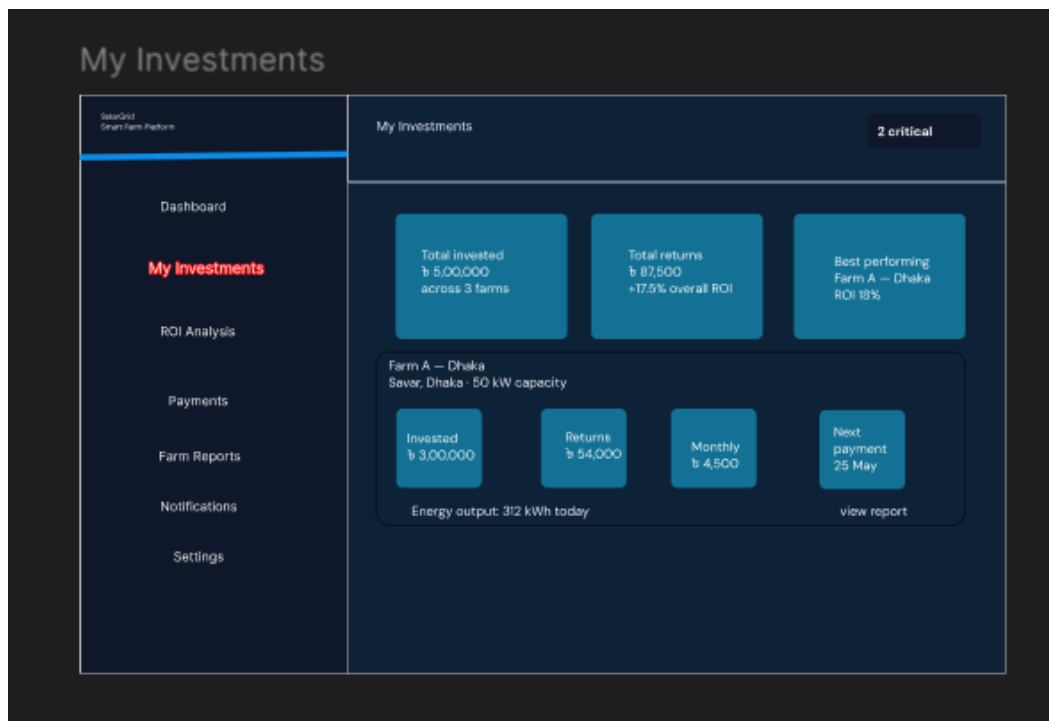


Fig 21: Investment Opportunities Interface.

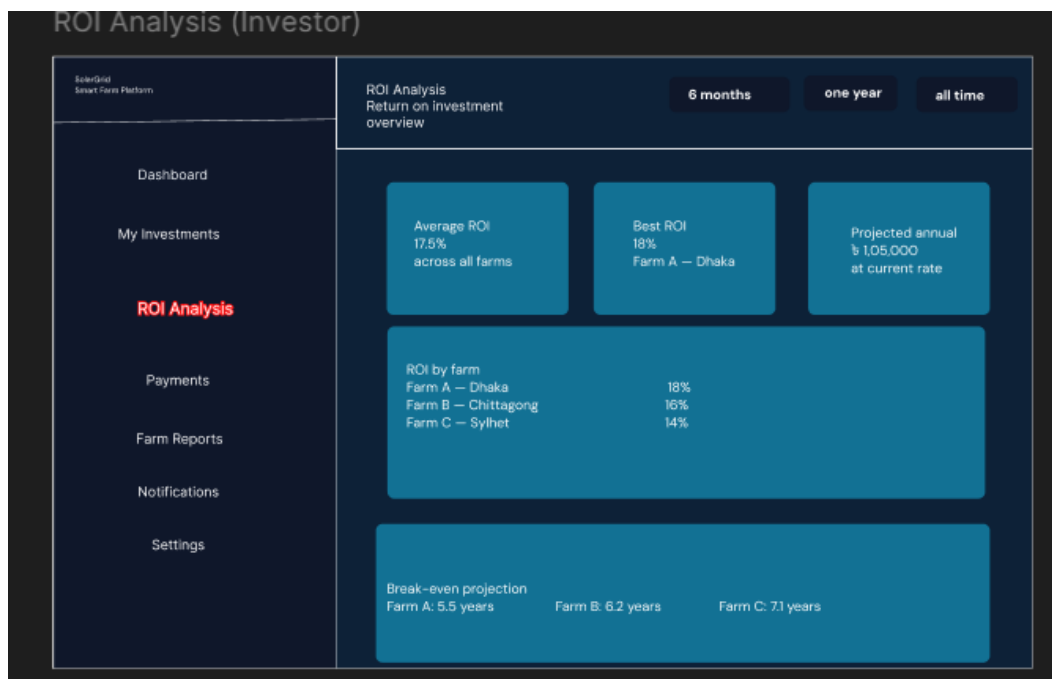


Fig 22: Financial Analysis Dashboard.

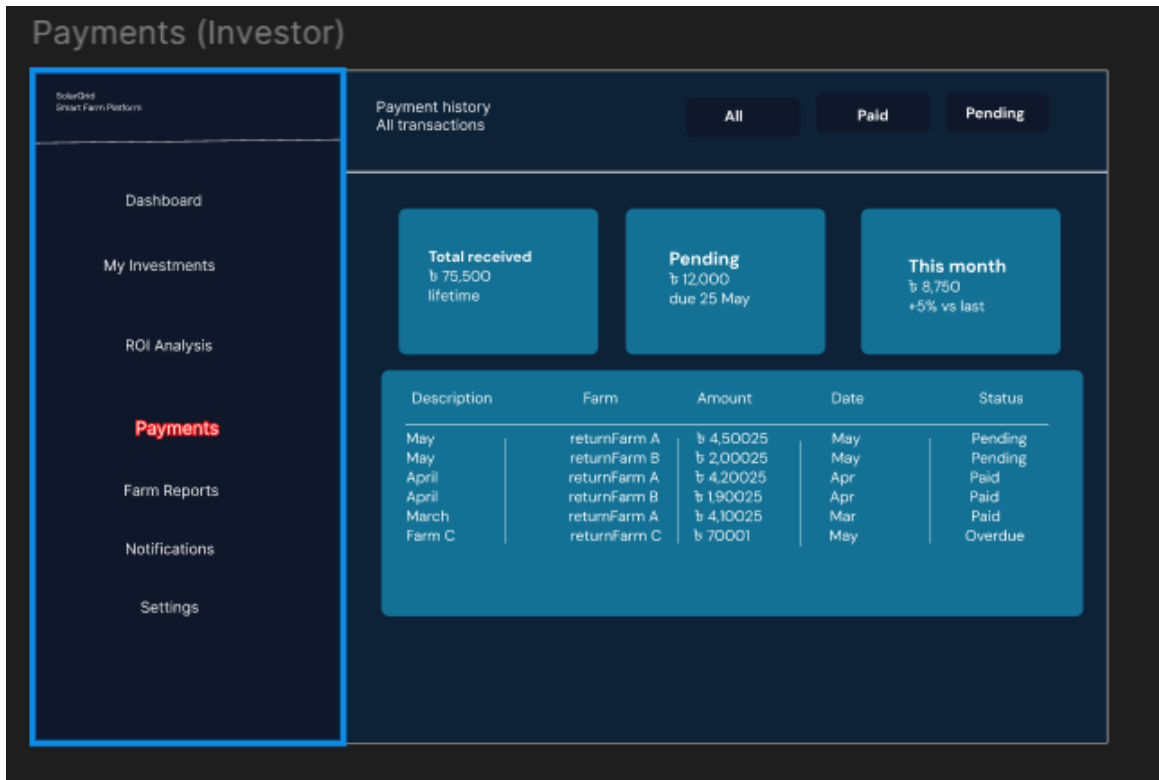


Fig 23: Payment Monitoring Interface.



Fig 24: Report.

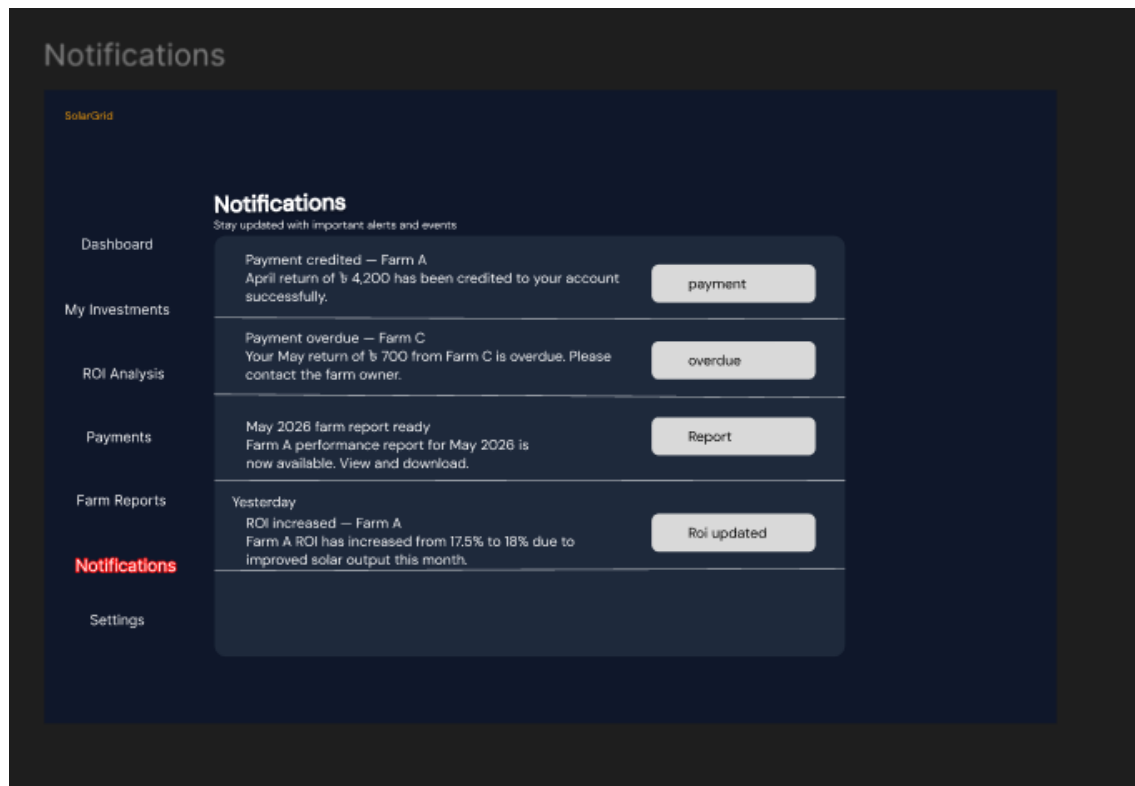


Fig 25: Notification.

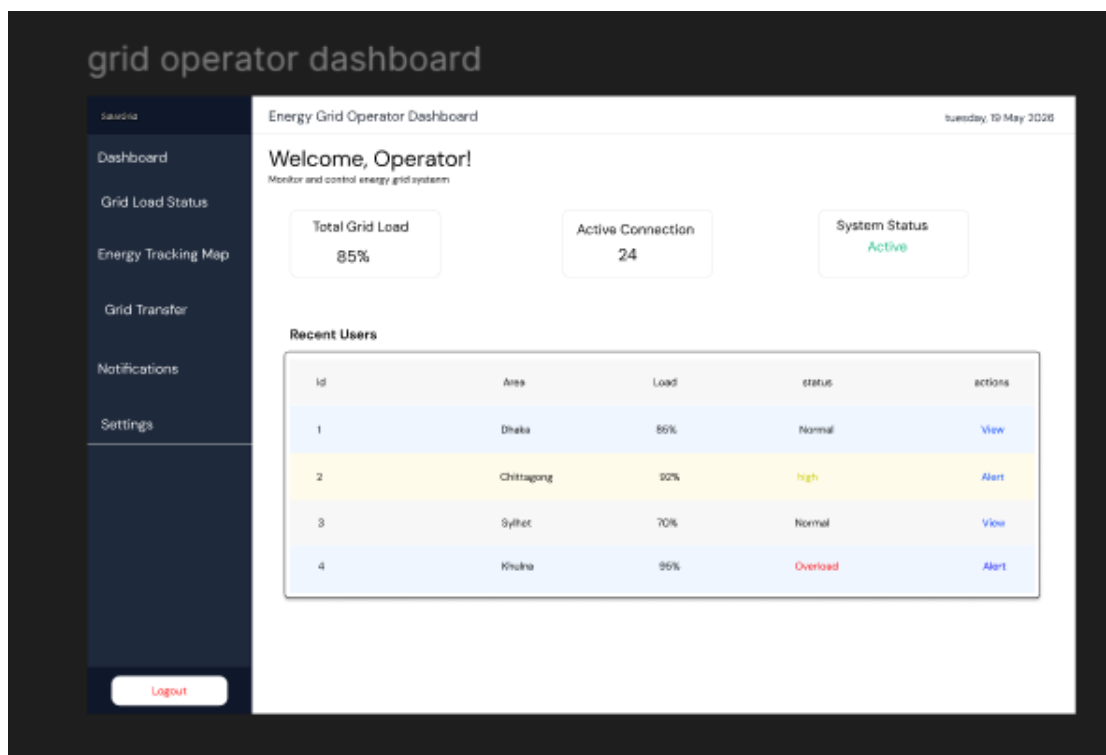


Fig 26: Grid Operator Dashboard.

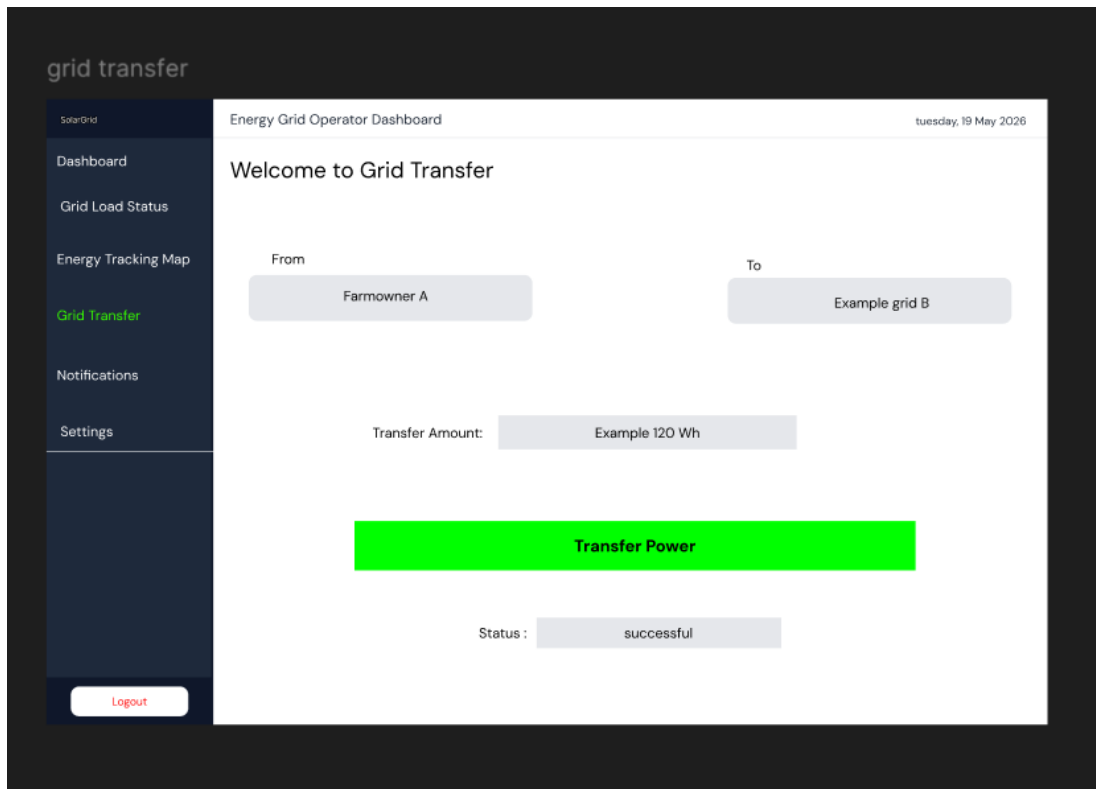


Fig 27: Grid Transfer.

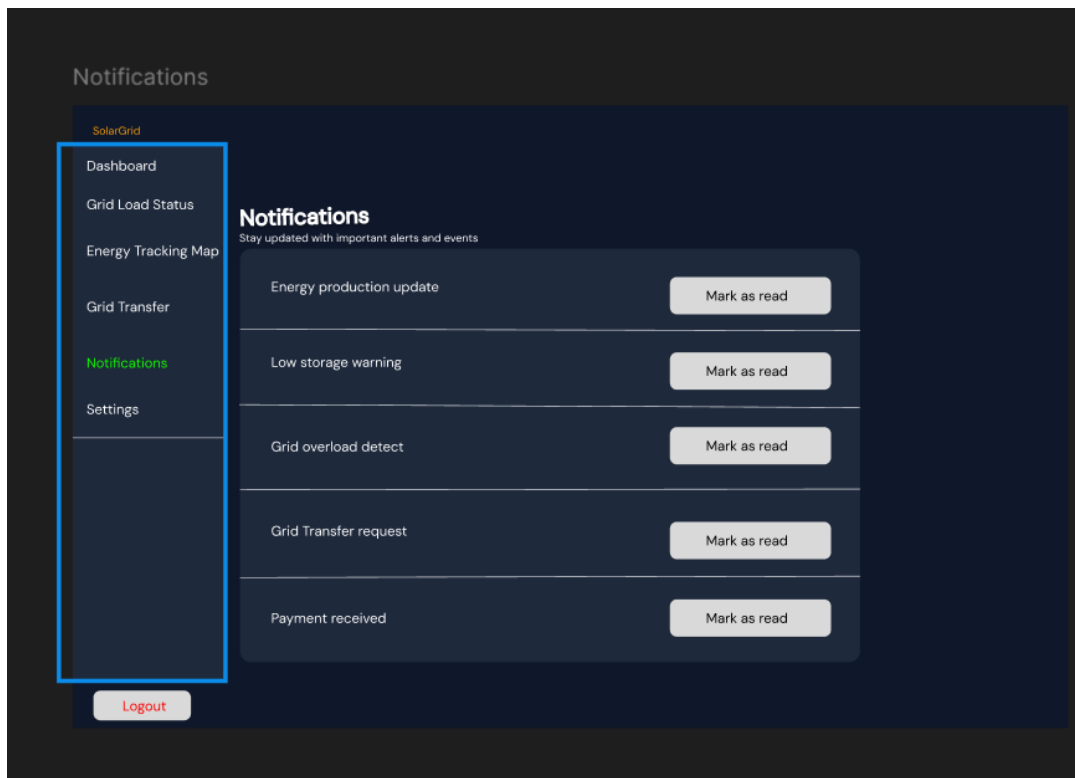


Fig 28: notifications.

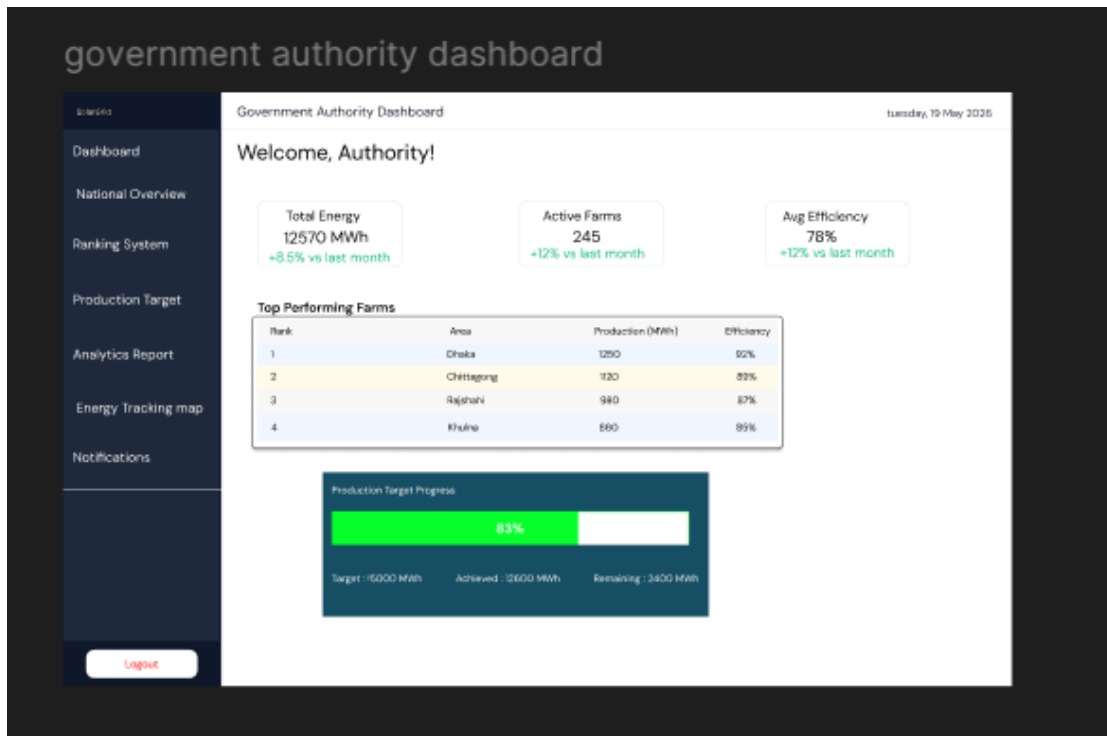


Fig 29: Government Authority Dashboard.

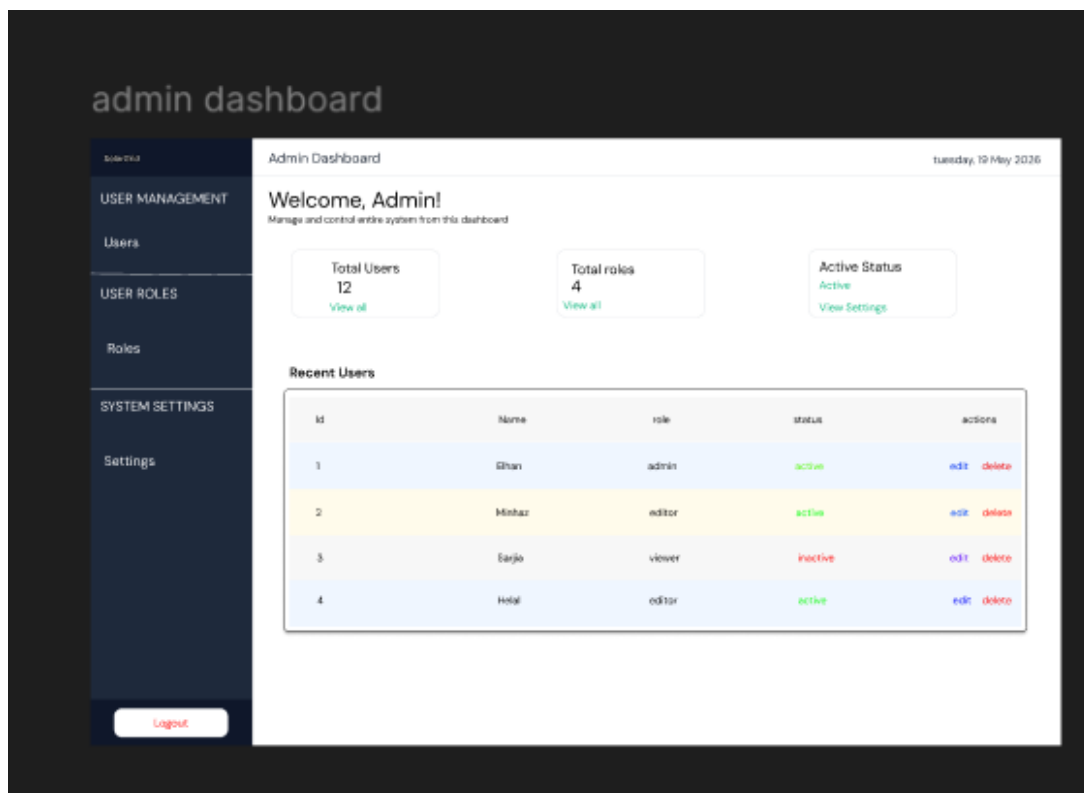


Fig 30: Administrative Control Dashboard.

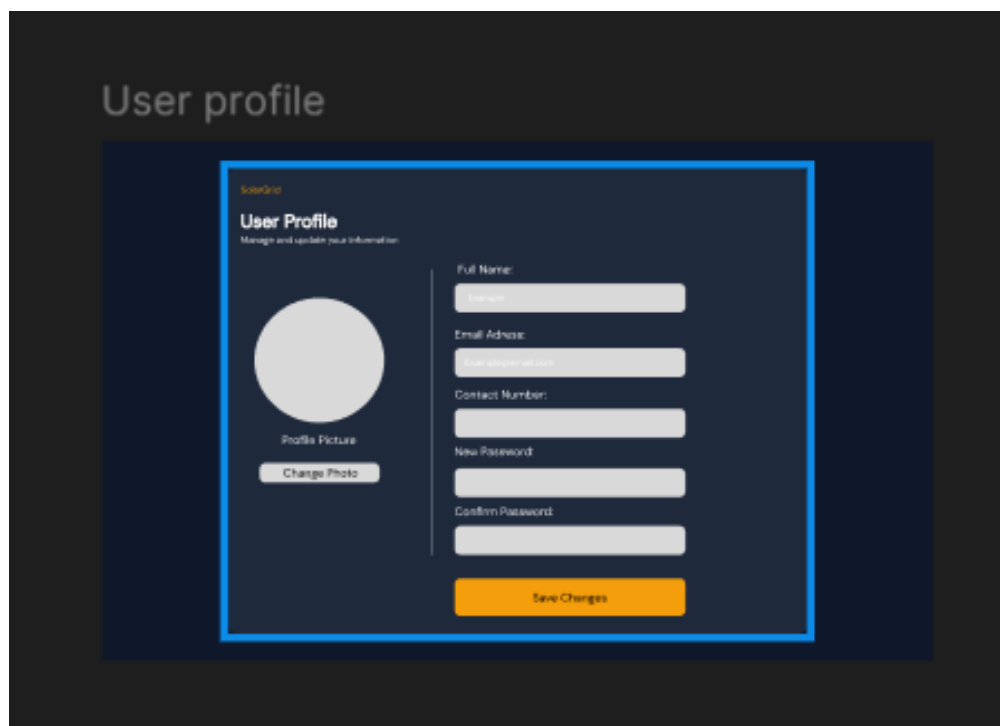


Fig 31: User Profile.

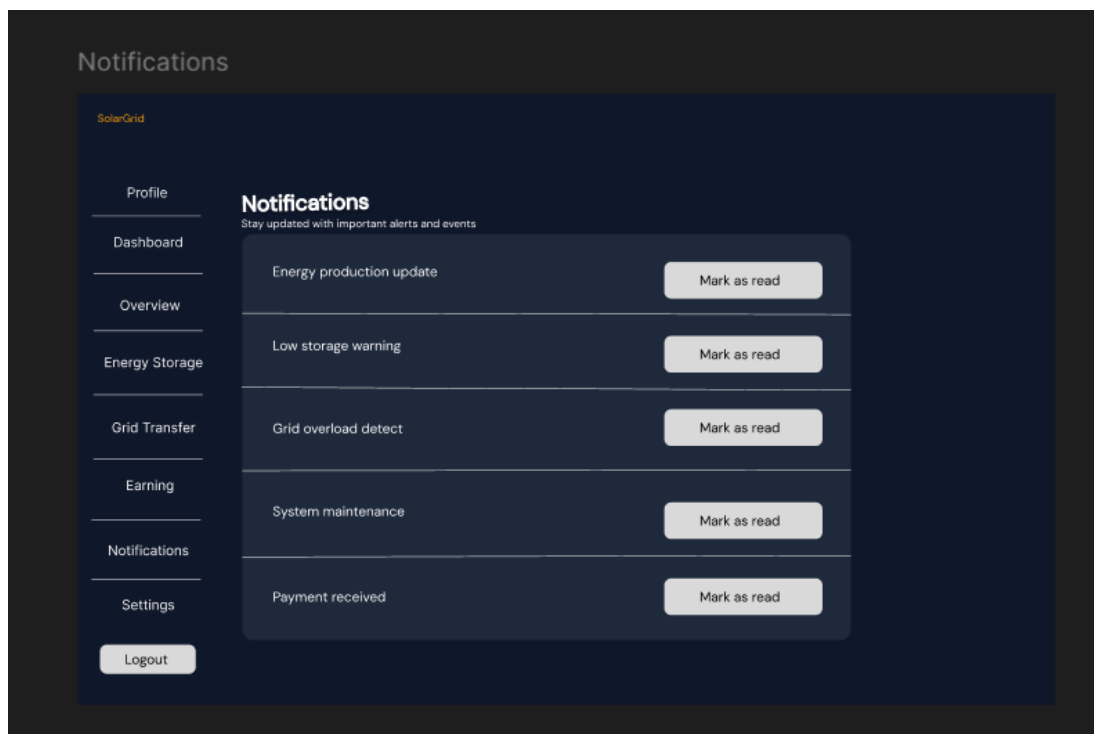


Fig 32: Notifications.

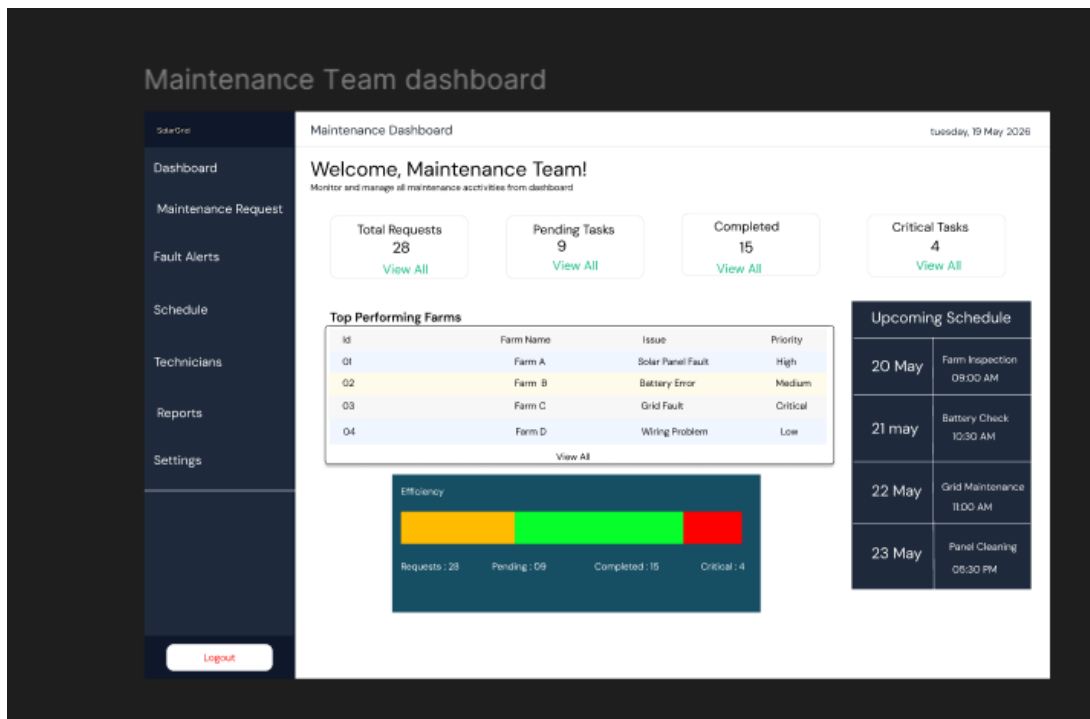


Fig 33: Maintenance Team dashboard.

5. GIT WORKFLOW

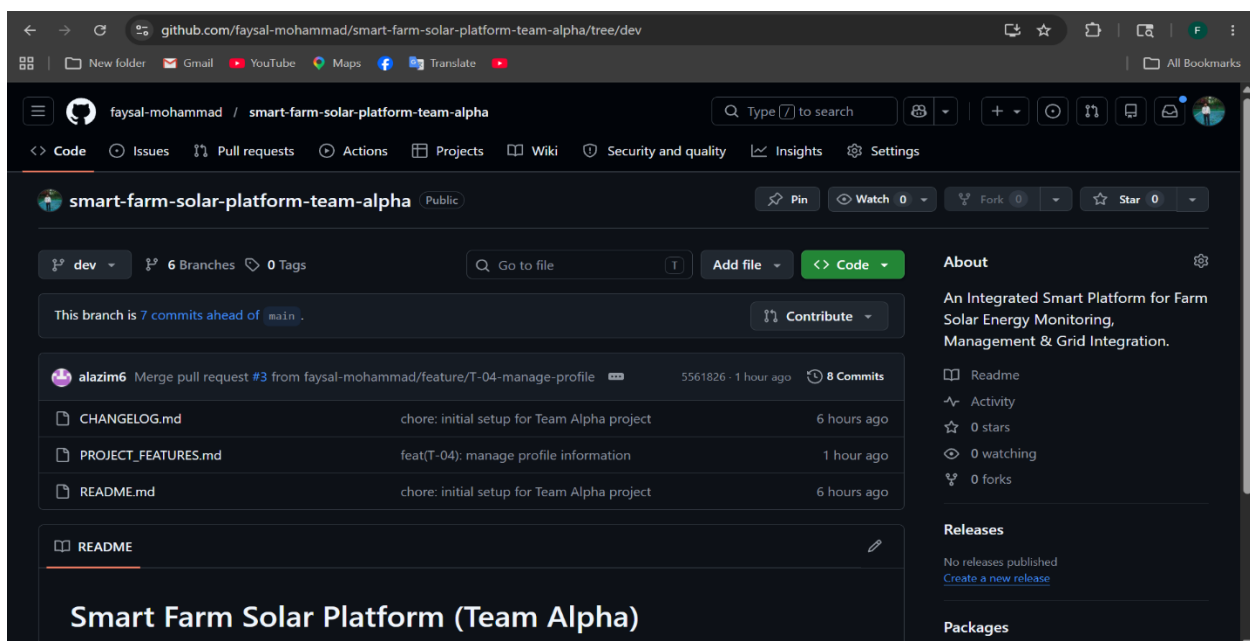


Fig 34: Git repository .

```
MINGW64/c/Users/faysa/smart-farm-solar-platform-team-alpha
faysa@faysal-acer MINGW64 ~ (main)
$ mkdir smart-farm-solar-platform-team-alpha
faysa@faysal-acer MINGW64 ~ (main)
$ cd smart-farm-solar-platform-team-alpha
faysa@faysal-acer MINGW64 ~/smart-farm-solar-platform-team-alpha (main)
$ git init
Initialized empty Git repository in C:/Users/faysa/smart-farm-solar-platform-team-alpha/.git/
faysa@faysal-acer MINGW64 ~/smart-farm-solar-platform-team-alpha (main)
$ echo "# Smart Farm Solar Platform (Team Alpha)" > README.md
faysa@faysal-acer MINGW64 ~/smart-farm-solar-platform-team-alpha (main)
$ echo "# Changelog" > CHANGELOG.md
faysa@faysal-acer MINGW64 ~/smart-farm-solar-platform-team-alpha (main)
$ echo "# Implemented Features" > PROJECT_FEATURES.md
faysa@faysal-acer MINGW64 ~/smart-farm-solar-platform-team-alpha (main)
$ git add README.md CHANGELOG.md PROJECT_FEATURES.md
warning: in the working copy of 'CHANGELOG.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'PROJECT_FEATURES.md', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'README.md', LF will be replaced by CRLF the next time Git touches it
faysa@faysal-acer MINGW64 ~/smart-farm-solar-platform-team-alpha (main)
$ git commit -m "chore: initial setup for Team Alpha project"
[main (root-commit) c540e07] chore: initial setup for Team Alpha project
3 files changed, 3 insertions(+)
create mode 100644 CHANGELOG.md
create mode 100644 PROJECT_FEATURES.md
create mode 100644 README.md
faysa@faysal-acer MINGW64 ~/smart-farm-solar-platform-team-alpha (main)
$ git branch -M main
faysa@faysal-acer MINGW64 ~/smart-farm-solar-platform-team-alpha (main)
$ git branch dev
faysa@faysal-acer MINGW64 ~/smart-farm-solar-platform-team-alpha (main)
$ git branch stage
faysa@faysal-acer MINGW64 ~/smart-farm-solar-platform-team-alpha (main)
$ git branch
  dev
* main
  stage
faysa@faysal-acer MINGW64 ~/smart-farm-solar-platform-team-alpha (main)
```

Fig 35: Git bash.

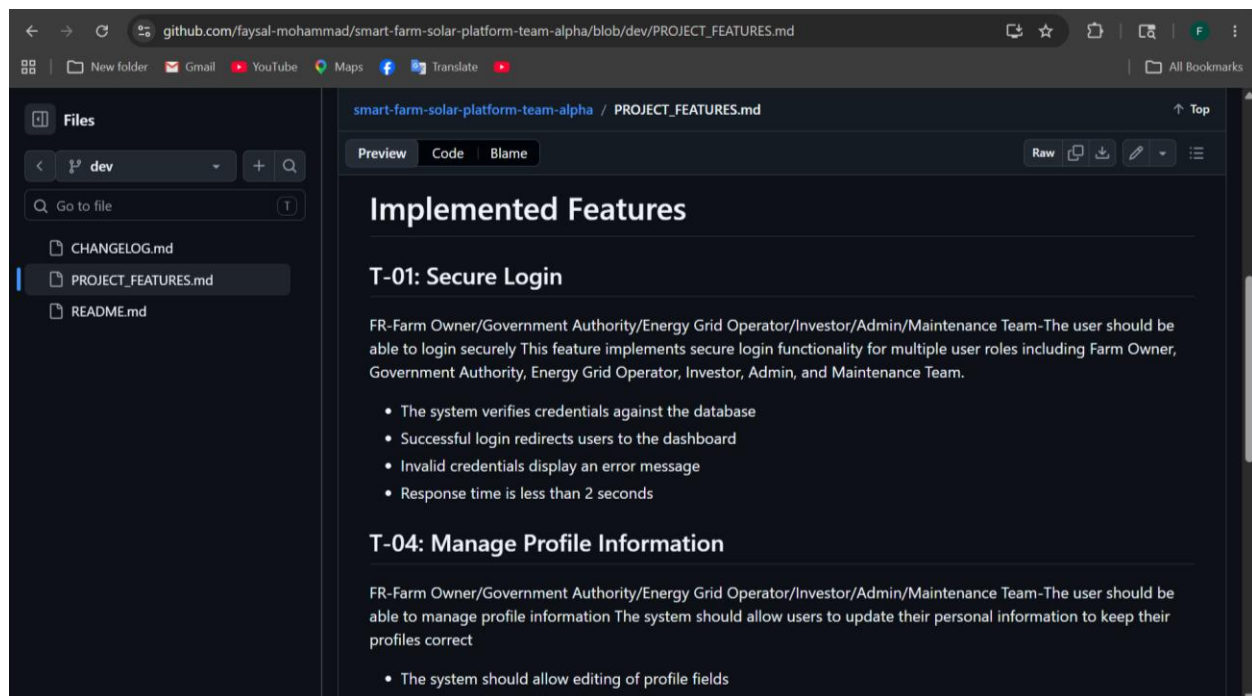


Fig 36: Branch view.

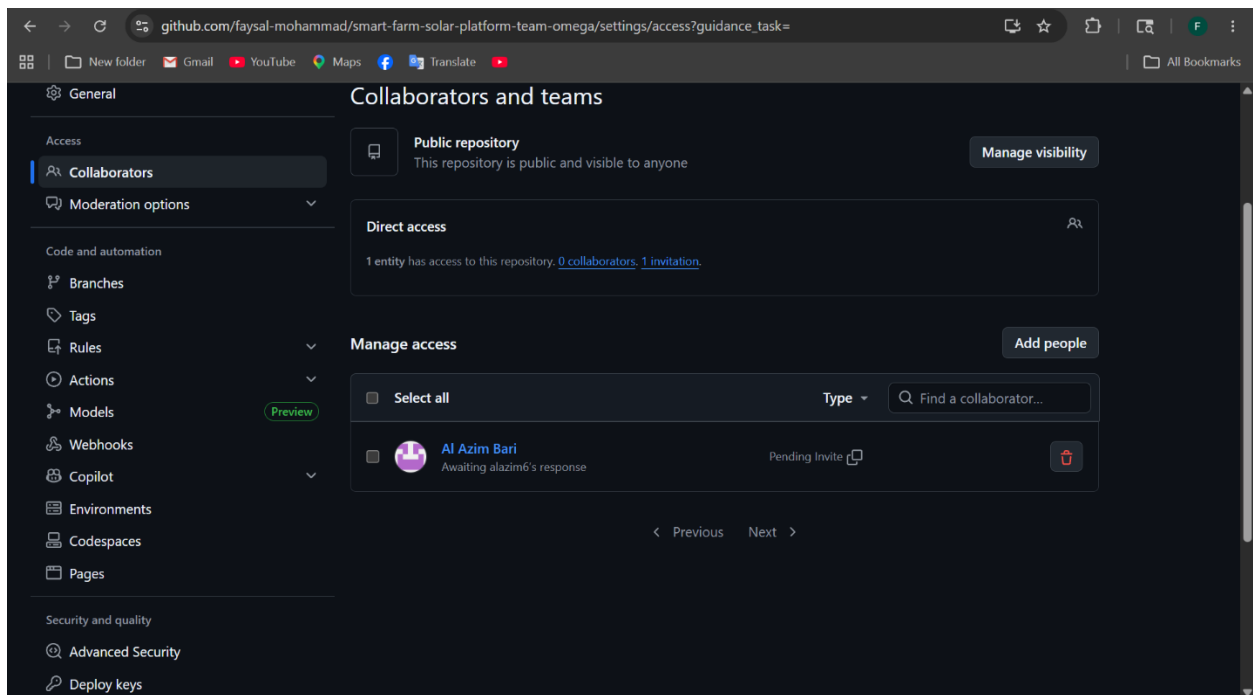


Fig 37: Git

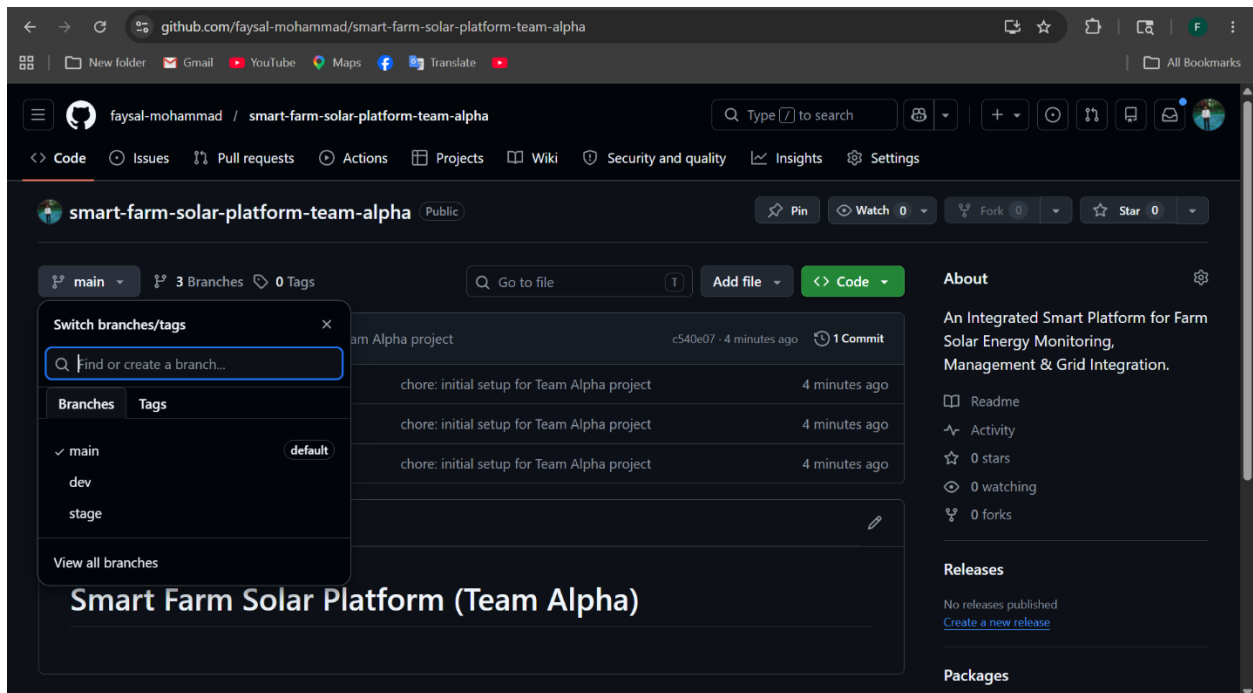


Fig 38.1: Git

Trello Board:

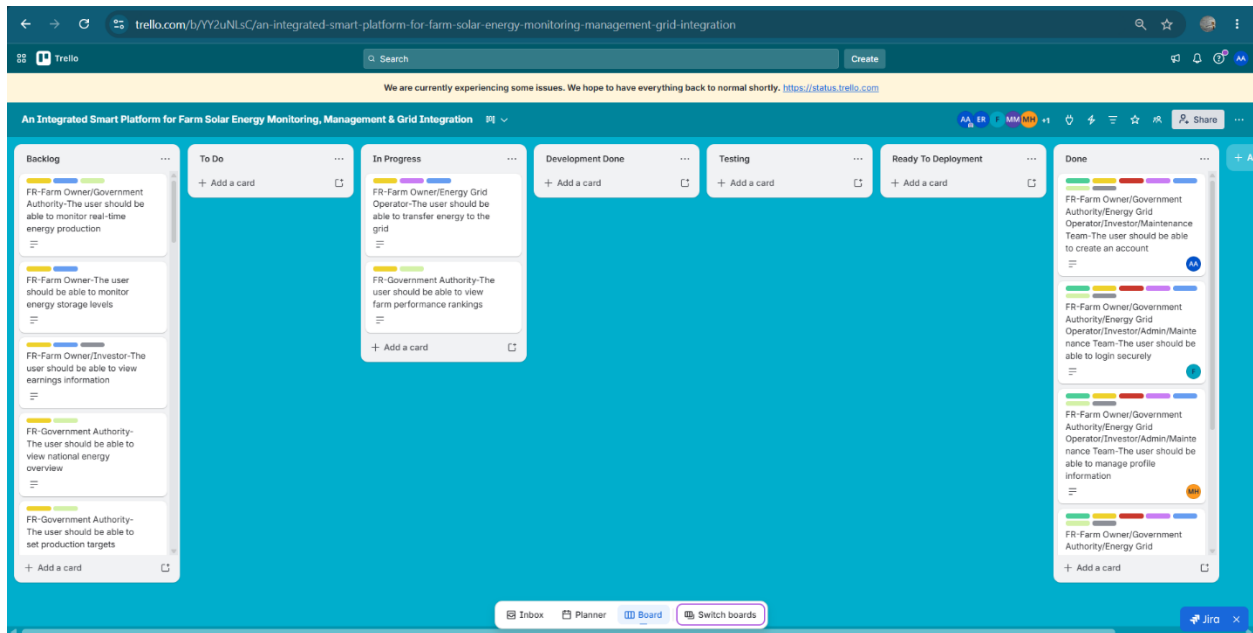


Fig 38.2: Trello Board.

6. SOFTWARE TESTING

For this project, the following testing methods should be used during the software testing phase:

Unit Testing

Unit testing should be used to test individual modules such as registration, login, payment monitoring, energy storage, and maintenance request separately.

Integration Testing

Integration testing should be used to verify whether connected modules work together correctly, such as login with dashboard, grid transfer with payment system, and maintenance request with notification system.

Smoke Testing

Smoke testing should be performed to ensure that the main functionalities of the system should work properly before detailed testing begins.

Black-box Testing

Black-box testing should be used to test the system based on functional requirements without considering the internal code structure.

System Testing

System testing should be used to test the complete smart farm solar platform as an integrated system.

Security Testing

Security testing should be used to verify secure login, role-based access control, session timeout, and protection against unauthorized access.

Performance Testing

Performance testing should be used to check system response time, real-time data updates, dashboard loading speed, and graph loading performance.

Recovery Testing

Recovery testing should be used to verify whether the system should recover properly after failure without losing important data.

Test Case

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.			Test Designed by: Azim	
Test Case ID: TC_01			Test Designed Date:1/5/26	
Test Priority (Low, Medium, High): High			Test Executed by: Azim	
Module Name: Registration			Test Execution Date:	
Test Title: Verify that a new user should be able to create an account with valid information.				
Description: This test checks whether a user should successfully register by entering valid name, email, password, confirm password, and selecting a role.				
Precondition: User should be on the registration page				
Dependencies: Internet connection and database should be available				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1.Go to registration page 2.Enter full name 3.Enter email 4. Enter password	Full Name: Al Azim Bari Email: alazimbari@gmail.com Password:abc@1234 Confirm Password: abc@1234	- Registration page should load successfully. - All required fields should be validated.		

5. Confirm password 6. Select role 7. Click Signup button	Role: Farm Owner/Investor/Energy Grid Operator/Government/Maintenance Team	- Email should be unique. - Password should be at least 8 characters. - Password and confirm password should match. - Account should be created within 30 seconds. - User should be redirected to the dashboard - No duplicate account should be created		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Azim		
Test Case ID: TC 02		Test Designed Date: 1/5/26		
Test Priority (Low, Medium, High): High		Test Executed by: Azim		
Module Name: Login		Test Execution Date:		
Test Title: Verify that a registered user should log in with valid credentials				
Description: This test checks whether the system allows a registered user to access the dashboard after entering a valid email and password.				
Precondition: User should have a registered account.				
Dependencies: Authentication database should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to login page 2. Enter email 3. Enter	Email: alazimbhari@gmail.com Password: abc@1234	- Login page should be loaded successfully.		

password 4. Click Login button		- Email and password should be verified. - Valid users should be redirected to dashboard. - Invalid credentials should show error message. - System response time should be less than 2 seconds. - Dashboard should display user-specific data		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Azim		
Test Case ID: TC_03		Test Designed Date: 2/5/26		
Test Priority (Low, Medium, High): High		Test Executed by: Azim		
Module Name: Forgot Password		Test Execution Date:		
Test Title: Verify that a user should reset password using a valid email				
Description: This test checks whether the system sends a secure reset link when the user enters a registered email address.				
Precondition: User should be on the forgot password page.				
Dependencies: Registered email and email service should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to forgot password page 2. Enter registered email 3. Click Send Reset Link button	Email: alazimbary@gmail.com	- Forgot password page should load successfully. - Email should be validated.		

		- System should verify email existence. - Reset link should be sent within 30 seconds. - Reset link should be secure and time-limited. - User should be able to set new password. - User should be redirected to the login page		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Azim		
Test Case ID: TC_04		Test Designed Date: 2/5/26		
Test Priority (Low, Medium, High): Medium		Test Executed by: Azim		
Module Name: Profile Management		Test Execution Date:		
Test Title: Verify that a user should update profile information successfully				
Description: This test checks whether users should edit and save their name, contact information, and password from the profile screen.				
Precondition: User should be logged in and on the profile page.				
Dependencies: User profile data should be available				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to profile page 2. Edit name 3. Edit contact 4.Address 5. Click save changes	Name: Al Azim Bari Contact: 017XXXXXXXXX Address: Block B	- Profile page should load successfully. - User should be able to edit profile information. - Input data should be validated.		

		- Changes should be saved successfully. - Updated data should be reflected immediately. - Confirmation messages should be displayed. - No data loss should occur		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Azim		
Test Case ID: TC_05		Test Designed Date: 3/5/26		
Test Priority (Low, Medium, High): Medium		Test Executed by: Azim		
Module Name: Notification		Test Execution Date:		
Test Title: Verify that users receive important system notifications				
Description: This test checks whether the system shows notifications instantly with message, type and time.				
Precondition: User should be logged in and notification feature should be active.				
Dependencies: Notification service should be available				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to notifications page 2. Check notification list 3. Click mark as read	Notification Type: Info/Warning Message: Low storage alert	- Notification should appear with message, type, and timestamp. - Notification should be categorized correctly. - Notification should appear within 2 seconds. - Notification should be marked as read successfully.		

		- Notification count should update correctly		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.			Test Designed by: Minhazul	
Test Case ID: TC_6			Test Designed Date: 3/5/26	
Test Priority (Low, Medium, High): High			Test Executed by: Minhazul	
Module Name: Real-Time Energy			Test Execution Date:	
Test Title: Verify that live energy production data should be displayed correctly				
Description: This test checks whether farm owners and Government Authority should view live energy production data on the energy overview screen.				
Precondition: User should be logged in and energy data should be available.				
Dependencies: Energy monitoring data sources should be connected.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to Energy Overview page 2. Check live energy graph 3. Select time filter	Time Filter: Daily/Weekly/Monthly/ Yearly	- Energy overview page should load successfully. - Live energy data should be displayed. - Graph should load within 2 seconds. - Data should be updated every 5 seconds. - Data accuracy should be at least 95%. - Time filters should work correctly. - Data should remain consistent across refresh		

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Minhazul		
Test Case ID: TC 7		Test Designed Date: 4/5/26		
Test Priority (Low, Medium, High): High		Test Executed by: Minhazul		
Module Name: Energy Storage		Test Execution Date:		
Test Title: Verify that stored energy level should be shown correctly				
Description: This test checks whether the system displays battery percentage and storage usage graph correctly.				
Precondition: User should be logged in and storage data should be available.				
Dependencies: Battery/storage monitoring data should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to energy storage page 2. Check battery percentage 3. Check storage graph	Battery Level: 0-100%	- Energy storage page should be loaded successfully. - Battery percentage should be accurate. - Storage graphs should be displayed. - Data should refresh within 5 seconds. - Low storage alert should trigger below threshold. - Battery level should remain consistent after refresh		

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Minhazul		
Test Case ID: TC 8		Test Designed Date: 4/5/26		
Test Priority (Low, Medium, High): High		Test Executed by: Minhazul		
Module Name: Grid Transfer		Test Execution Date:		

Test Title: Verify that a farm owner should submit energy transfer requests to the grid				
Description: This test checks whether a farm owner should enter energy amount, select grid connection and submit a transfer request.				
Precondition: Farm owners should be logged in.				
Dependencies: Grid connection and transfer service should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to grid transfer page 2. Enter energy amount 3. Select grid connection 4. Click submit transfer	Energy Amount: 50 kWh Grid Connection: National Grid	- Grid transfer pages should load successfully. - Energy amounts should be validated and accepted. - Grid connections should be selected successfully. - Transfer request should be submitted successfully. - Approval/rejection workflow should function correctly. - Transfer status should be updated in real-time. - Error messages should appear for failed transactions. - Transfer confirmation message should be displayed		

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.	Test Designed by: Minhazul
Test Case ID: TC_9	Test Designed Date: 4/5/26
Test Priority (Low, Medium, High): Medium	Test Executed by: Minhazul
Module Name: Earnings	Test Execution Date:

Test Title: Verify that users should view their earnings information				
Description: This test checks whether farm owners and investors should view total earnings and income graph correctly.				
Precondition: User should be logged in and earnings data should exist.				
Dependencies: Payment and earning calculation data should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to earnings page 2. Check total earnings 3. Apply time filter	Time Filter: Monthly/Yearly	- Earnings page should load successfully. - Total earnings should be displayed correctly. - Income graph should be shown. - Data should be updated in real time. - Time filters should work correctly. - Earnings data should match transaction records		

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Minhazul		
Test Case ID: TC 10		Test Designed Date: 5/5/26		
Test Priority (Low, Medium, High): High		Test Executed by: Minhazul		
Module Name: Logout		Test Execution Date:		
Test Title: Verify that a user should log out successfully from the system.				
Description: This test checks whether a logged-in user should safely log out from the system and return to the login page.				
Precondition: User should be logged into the system.				
Dependencies: Internet connection and active user sessions should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Go to dashboard 2. Click Logout button	Logged-in user account	- Dashboard should be visible before logout. - Logout should be successful. - User session should be terminated securely. - User should be redirected to login page. - User should not access system without logging in again. - User should not access protected pages after logout		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Faysal		
Test Case ID: TC 11		Test Designed Date: 5/5/26		
Test Priority (Low, Medium, High): High		Test Executed by: Faysal		
Module Name: National Overview		Test Execution Date:		
Test Title: Verify that Government Authority should view national energy data				
Description: This test checks whether Government Authority should view total energy production, energy distribution maps and country statistics.				
Precondition: Government Authority should be logged in.				
Dependencies: National energy data and map service should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Government Authority 2. Go to National Overview page 3. Select region filter	Region: Dhaka	- National overview page should load successfully. - Total energy chart should display correctly.		

		- Energy distribution maps should be visible. - Region filter should work correctly. - Data should be accurate and up to date.		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Faysal		
Test Case ID: TC 12		Test Designed Date: 5/5/26		
Test Priority (Low, Medium, High): Medium		Test Executed by: Faysal		
Module Name: Production Target		Test Execution Date:		
Test Title: Verify that Government Authority should set production targets				
Description: This test checks whether authorized Government Authority should enter and save energy production targets for farms.				
Precondition: Government Authority should be logged in.				
Dependencies: Production target module should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Government Authority 2. Go to production target page 3. Enter target value 4. Click save target	Target Value: 5000 kWh	- Production target page should load successfully. - Target value should be accepted. - Target should be saved successfully. - Target should be visible to Farm Owners. - Only authorized users should edit targets.		

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Faysal		
Test Case ID: TC 13		Test Designed Date: 6/5/26		
Test Priority (Low, Medium, High): Medium		Test Executed by: Faysal		
Module Name: Analytics Report		Test Execution Date:		
Test Title: Verify that analytics reports should be generated and exported				
Description: This test checks whether Government Authority should generate performance reports and export them successfully.				
Precondition: User should be logged in and report data should exist.				
Dependencies: Analytics data and export service should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to analytics report page 2. Select report period 3. Click export report	Period: Monthly/Yearly Export Type: PDF/Word	- Analytics report page should be loaded successfully. - Reports should be generated correctly. - Performance data should be displayed. - Export option should work successfully. - Report should be downloaded within 3 seconds.		

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Faysal		
Test Case ID: TC 14		Test Designed Date: 6/5/26		
Test Priority (Low, Medium, High): Low		Test Executed by: Faysal		
Module Name: Reward System		Test Execution Date:		
Test Title: Verify that rewards should be assigned to eligible users				

Description: This test checks whether eligible farm owners should be selected and assigned rewards from the reward management screen.				
Precondition: Authorized user should be logged in.				
Dependencies: Data records and export service should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to reward management page 2. Select eligible user 3. Enter reward details 4. Click assign reward	User: Farm Owner Reward: Energy Bonus	- Reward management page should load successfully. - Eligible users should be listed. - Reward details should be accepted. - Reward should be assigned correctly. - User should receive notification.		

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.			Test Designed by: Faysal	
Test Case ID: TC_15			Test Designed Date: 6/5/26	
Test Priority (Low, Medium, High): High			Test Executed by: Faysal	
Module Name: Admin Panel			Test Execution Date:	
Test Title: Verify that admin should access and manage the system dashboard				
Description: This test checks whether the admin should access user, role and system setting controls from the admin dashboard.				
Precondition: Admin should be logged in.				
Dependencies: Admin permission and dashboard service should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Login as admin 2. Go to admin dashboard 3. Open user, role, and settings menu	Role: Admin	- Admin dashboard should load successfully. - User, role, and setting controls should be visible. - Admin should be able to access all control functions. - System management operations should work correctly.		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Elhan		
Test Case ID: TC 16		Test Designed Date: 7/5/26		
Test Priority (Low, Medium, High): Medium		Test Executed by: Elhan		
Module Name: Energy Tracking Map		Test Execution Date:		
Test Title: Verify that energy flow map displays farm locations and energy direction				
Description: This test checks whether the map shows energy flow, farm markers and flow direction clearly.				
Precondition: Energy Grid Operator/ Government Authority should be logged in.				
Dependencies: Map service and energy flow data should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to energy flow map 2. Check farm markers 3. Check energy flow direction 4. Click refresh	Map Layer: Energy Flow	- Energy map should load successfully. - Farm locations should be displayed. - Energy flow direction should be visible clearly. - Map should update in real-time. - Data should refresh within 5 seconds.		

		- Map should load without delay		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Elhan		
Test Case ID: TC_17		Test Designed Date: 7/5/26		
Test Priority (Low, Medium, High): High		Test Executed by: Elhan		
Module Name: Grid Load Status		Test Execution Date:		
Test Title: Verify that overload warning appears when grid load is high				
Description: This test checks whether the system shows grid load status and triggers an overload warning when the load crosses the threshold.				
Precondition: Energy Grid Operator should be logged in.				
Dependencies: Grid load data and alert system should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to grid load status page 2. Check current load 3. Simulate high load condition	Grid Load: Above threshold	- Grid load page should load successfully. - Load graph should be displayed correctly. - Current load should be shown in real-time. - Load threshold should be visible. - Overload alert should trigger correctly.		

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Elhan		
Test Case ID: TC_18		Test Designed Date: 7/5/26		
Test Priority (Low, Medium, High): High		Test Executed by: Elhan		
Module Name: Grid Request		Test Execution Date:		
Test Title: Verify that a farm owner should submit a grid connection request.				

Description: This test checks whether a farm owner should submit a grid connection request to sell energy to the grid.				
Precondition: Farm owners should be logged in to the system.				
Dependencies: Grid request form and approval system should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to Grid Connection Request page 2. Fill request form 3. Click submit request button	Farm ID: FARM-101 Request Type: Grid Connection	- Grid request page should load successfully. - Input data should be accepted. - Request should be submitted successfully. - Approval/rejection workflow should function correctly. - Request status should be displayed.		

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Elhan		
Test Case ID:TC 19		Test Designed Date: 8/5/26		
Test Priority (Low, Medium, High): Medium		Test Executed by: Elhan		
Module Name: Financial Analysis		Test Execution Date:		
Test Title: Verify that ROI analysis is calculated based on investment amount				
Description: This test checks whether investors should enter an investment amount and view expected return through the ROI graph and summary.				
Precondition: Investor should be logged in.				
Dependencies: Financial data and ROI calculation module should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to financial analysis page 2. Enter investment amount	Investment Amount: 100000 BDT	- Financial analysis page should load successfully.		

3. Check ROI result		- Investment amount should be accepted. - ROI graph should be displayed. - Expected return should be calculated accurately. - Data should be updated dynamically.		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Elhan		
Test Case ID:TC-20		Test Designed Date: 8/5/26		
Test Priority (Low, Medium, High): Medium		Test Executed by: Elhan		
Module Name: Payment Monitoring		Test Execution Date:		
Test Title: Verify that payment history should be displayed correctly				
Description: This test checks whether farm owners, investors and admins should view payment records with date, amount and status.				
Precondition: Farm Owner / Investor / Admin should be logged in.				
Dependencies: Payment records should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to payment monitoring page 2. View payment table 3. Use search/filter option	Search: Paid Filter: Monthly	- Payment monitoring page should be loaded successfully. - Payment records should be displayed correctly. - Date, amount, and status should be accurate. - Search and filter should work properly. - Data should update instantly.		

		- Payment records should match stored transaction data		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Sarjia		
Test Case ID:TC 21		Test Designed Date: 9/5/26		
Test Priority (Low, Medium, High): Low		Test Executed by: Sarjia		
Module Name: Energy Sharing		Test Execution Date:		
Test Title: Verify that a farm owner should share energy with nearby farms.				
Description: This test checks whether a farm owner should enter energy amount, select a nearby farm and share energy successfully.				
Precondition: Farm owners should be logged in and have available surplus energy.				
Dependencies: Nearby farm list and energy sharing service should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to Energy Sharing page 2. Enter energy amount 3. Select recipient farm 4. Click Share Energy button	Energy Amount: 20 kWh Recipient: Nearby Farm A	- Energy sharing pages should be loaded successfully. - Energy amounts should be validated and accepted. - Recipient should be selected successfully. - Energy sharing should be completed successfully. - Sharing status should be updated.		

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Sarjia		
Test Case ID:TC 22		Test Designed Date: 9/5/26		
Test Priority (Low, Medium, High): Medium		Test Executed by: Sarjia		
Module Name: Maintenance Request		Test Execution Date:		
Test Title: Verify that farm owners should submit maintenance requests				
Description: This test checks whether a farm owner should submit an issue description and image for maintenance support.				
Precondition: Farm owners should be logged in.				
Dependencies: Maintenance request form and image upload service should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to maintenance request page 2. Enter issue description 3. Upload image 4. Click submit request	Description: Solar panel not working Image: panel_issue.jpg	- Maintenance request page should be loaded successfully. - Description and image should be accepted. - Request should be submitted successfully. - Request status should be trackable.		

Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Sarjia		
Test Case ID:TC_23		Test Designed Date: 9/5/26		
Test Priority (Low, Medium, High): High		Test Executed by: Sarjia		
Module Name: Role Management		Test Execution Date:		
Test Title: Verify that admin should assign or update user roles successfully.				
Description: This test checks whether an admin should select a user, assign a role, and update access permission successfully.				
Precondition: Admin should be logged into the system.				
Dependencies: User list and role assignment option should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)

1. Go to Role Management page 2. Select user 3. Select new role 4. Click update role button	User: alazimbari@gmail.com Role: Energy Grid Operator	- Role management pages should load successfully. - User selection should work correctly. - Role should be updated successfully. - Changes should be applied immediately. - Access permissions should be updated correctly.		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Sarjia		
Test Case ID:TC 24		Test Designed Date: 10/5/26		
Test Priority (Low, Medium, High): High		Test Executed by: Sarjia		
Module Name: Login Attempt Limit		Test Execution Date:		
Test Title: Verify that the system blocks login after repeated failed attempts				
Description: This test checks whether the system blocks or restricts access after the allowed number of failed login attempts.				
Precondition: User should be on the login page.				
Dependencies: Authentication system and login attempt counter should be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Go to login page 2. Enter wrong password multiple times 3. Try to login again after limit	Email: alazimbari@gmail.com Wrong Password: wrong123 Attempts: 3	- Invalid login error should be displayed. - System should count failed attempts. - Account should be blocked after 3 failed attempts.		

		- User should receive warning/notification.		
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Project Name: An Integrated Smart Platform for Farm Solar Energy Monitoring, Management & Grid Integration.		Test Designed by: Sarjia		
Test Case ID:TC 25		Test Designed Date: 10/5/26		
Test Priority (Low, Medium, High): Medium		Test Executed by: Sarjia		
Module Name: Low Internet Support		Test Execution Date:		
Test Title: Verify that the system remains usable under low internet connection				
Description: This test checks whether the system keeps important features usable and reduces unnecessary data loading under low bandwidth.				
Precondition: User should access the system under low internet condition.				
Dependencies: Low bandwidth network conditions should be available or simulated.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open system under low internet 2. Go to dashboard 3. Open important features 4. Check loading behavior	Network: Low bandwidth	- System should remain usable under low bandwidth. - Important features should be loaded successfully. - Unnecessary data loading should be reduced. - System should not crash or freeze. - Basic features should remain accessible.		

7. CONCLUSION

The “**Smart Farm Solar Energy Monitoring, Management & Grid Integration**” project is designed to help farm owners monitor and manage solar energy easily. The system includes features such as user registration, login, real-time energy monitoring, energy storage, grid energy transfer, payment monitoring, maintenance request, reports and admin control. Different users such as Farm Owner, Government Authority, Investor, Energy Grid Operator, Maintenance Team and Admin can use the system based on their roles.

In this project, we prepared the project proposal, selected the process model, created the Gantt chart, wrote functional and non-functional requirements, and designed use case and class diagrams. We also planned UI/wireframe design, Git workflow, Trello board and software testing. The semi-detached COCOMO model was used to estimate effort, development time and required people.

However, this project has some limitations because real-time monitoring depends on internet connection, sensors and proper hardware support. In the future, the system can be improved by adding a mobile app, IoT integration, AI-based prediction and advanced security features.

Overall, this project can help improve solar energy monitoring and management in farming. It can also support better energy use, grid connection, maintenance and decision-making.